

# Universe24: one integer rule on a cubic lattice, read through clicks, gives the wave equation, the quantum of action and a weak-field metric as three readings of one line

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## Abstract

We present one method of computing physics: a cubic lattice whose Nodes hold whole numbers, stepped by one integer recurrence and read only through the clicks of declared detectors. On it the classical wave equation, the quantum of action and the metric of gravity are three readings of one line, with nothing continuous beneath; every formula below is an example of what the method yields, with a list of what it does not yet give. The paper claims the method and not nature: it does not assert that the world is such a lattice, and every result is a supporting finding for the method, offered for the reader's test and use. The recurrence steps every Node from its present, its past and its six neighbours by one integer division, reversibly and covariantly, conserving a quadratic form. Measurement is the meeting of the record run forward with the one read backward from the click. Newton's gravitation is derived,  $G$  as a reading. Einstein's weak field is derived in the clock's potential under one named assumption, a Link clocked twice by its Node: the exponential metric,  $\beta = \gamma = 1$  in that potential, a shadow 4.6 percent above Schwarzschild's; against Kepler's potential the computed pair departs by 25 to 33 percent, a gap closing below  $10^{-18}$ . The three computed worlds are derivations, not experiments: the two-slit pattern is the line stepped on its world, and Bell's  $S = 2\sqrt{2}$  and GHZ's  $M = -4$  follow from the line's linearity under one declared credit, every local credit giving at most 2.

**Keywords:** integer lattice rule; reversible cellular automaton; measurement as a meeting of past and future; Bell's inequality; weak-field gravity; limiting speed on a lattice

## 1 Introduction

**The question.** The object of this paper is one method of computing physics: a GameBoard of Nodes stepped by one local, reversible rule in whole numbers, read only through clicks, the one measurement there is. On it the classical wave equation, the quantum of action and the metric of gravity are three readings of one line, Rule3: the line itself is the lattice wave equation, its count read in quanta is the click, and its paces read as a clock and a ruler are the metric, with nothing continuous beneath. The question is whether that method can give the clicks that nature shows, and with how little state, non-locality and randomness; the fifteen formulas and the three computed worlds below are examples of what the method yields, not its scope, and Section 10.3 lists what it does not yet give. The paper's one claim is the method: a GameBoard of integers stepped by one rule and read into the world through clicks, from which the forms of nature are derived in the order stated. It does not claim that nature is such a lattice; every derivation, comparison and prediction below is a supporting finding for the method, each with its status, and none is the paper's thesis. The judgement is the reader's: the law's document, the implementation and the files are public, and the paper is written to be used for further derivations and tests, as the papers it cites are used here. Of Newton the paper derives gravitation: Poisson's equation as a field's rest on the lattice, the inverse-square pull as a free packet's fall in the field's gradient at the band's inertia, the mass as the count, Kepler's orbits, and  $G$  as a reading of the law's constants. Of Einstein it derives the weak field in the clock's potential, the redshift, the bending, Shapiro's delay and the perihelion, with  $\gamma = \beta = 1$  under one named assumption, a Link clocked twice by its reading Node's own clock, and the special-relativistic part only as the matter family's gap closes, the moving clock's factor reaching Lorentz's as the family's kinetic scale reaches light's. The structure is a scalar pace and not the tensor field equations, so it departs from Einstein's at the second order and in the strong field, has no frame dragging and not the tensor's polarisations, and against the Newtonian potential the orbits measure the computed pair's departures are the band's gap, closing with it (Section 8). It starts from one statement and never contradicts it (ASSUMPTION): the world is the meeting of

the past with the future; a click is that meeting, the one measurement there is; and an observer sees only clicks (the words are defined at the end of this section). Every physical number of the paper is a reading of one integer rule with its declared integers, the pairs,  $\Gamma$ ,  $T$ , the two holders' weights, gravity's  $E_g$  and the sign holder's  $k$ , and the vacuum content (Section 6.6); the implementation holds no number of physics, no floating-point arithmetic and no random draw in the law; that these are fewer declared integers than the standard model's nineteen parameters is an explicitly named hypothesis a reader can check (HYPOTHESIS); of nature's other clicks it gives the forms of Section 8 at the computed pair, with the findings of Section 10.3 against them. A Node holds a record's present and its past, two integer levels, and the rule makes the record's future from the two and from the six neighbouring Nodes. The rule is covariant under the lattice's translations and under the symmetry group of the cube, the 24 rotations and, with the reflections, the 48 signed permutations of the three axes; no boost is among them (Section 3.1). The lattice holds nothing but events that split: at every Node a level splits by the rule to the six neighbours. Run forward from an emitter, the split events are the future, every path at once; read backward from a realised click by the same rule, they are the past, and through two slits both pass both gaps. A click is where the two meet at a declared detector, and which meeting is realised is the one thing the model does not determine, the one non-local element of the description (Section 5.2). What appears as a wave is the future, what appears as a particle is the present, the click, and the track is the past (Section 5.1). Least is meant in three named counts, the state per Node, the non-local elements and the random draws, among the models we know of; no theorem of minimality is claimed, and why nature is so we do not claim to know.

**The rule in one line.** Everything in the paper follows from one line of integer arithmetic, Rule3, together with one write and three readings. At every Node, in every interval and for every family of physical values [1, the line; the booking; the primitives]:

|                |                                                                                                                                                                                                                                                                                                                                                  |     |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Rule3:         | $w a_{\text{next}} + r' = \sum_{j \sim i} R_{ij} \text{arr}_j + S a_{\text{now}} - w a_{\text{before}} + r, \quad 0 \leq r' < w,$ $w = 6 \text{den } \Gamma^2, \quad R_{ij} = 2 \text{num } p_a(i, j)^2 = 2 \text{num } p_i^2 q_{ij}^2 / \Gamma^2,$ $S = 12 \text{den } \Gamma^2 - 12 (\text{den} - \text{num}) p_0^2 - \sum_{j \sim i} R_{ij};$ | (1) |
| the one write: | $a \text{ held part at a Node gains } \left( \sum_f w_f q_f(\text{Node}) + r \right) \text{div } E, \text{ the remainder } r \text{ kept there;}$                                                                                                                                                                                                |     |
| the readings:  | $D_i = \text{now}_i^2 - \text{next}_i \text{ before}_i, \quad W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}},$ $F_{ij} = \text{num} (\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j).$                                                                                 |     |

In the first line  $a_{\text{now}}$ ,  $a_{\text{before}}$  and  $a_{\text{next}}$  are the family's level at the Node now, one interval before and one interval later: the present, the past and the future.  $\text{arr}_j$  is the arrival through the Port to the neighbour  $j$ , that neighbour's level, the six Ports summed.  $r$  and  $r'$  are the remainder kept at the Node before and after the step,  $\Gamma$  the Node clock.  $[\text{num}, \text{den}]$  is the family's pair, and  $\cos \omega_0 = \text{num}/\text{den}$  is its rotation per interval at wave number zero.  $p_0$  is the Node's clock pace,  $p_i$  its own pace and  $p_a(i, j)$  the pace of the Link through the Port, the Node's pace times the Link's factor  $q_{ij}/\Gamma$ , read the same from both ends. All are the Node clock lowered by the field levels the family reads there (Section 3.4). At the vacuum's paces the two Links of an axis share one  $R_a = 2 \text{num } \Gamma^2$  and  $S = 0$ .  $\text{div}$  is the floor division. The second line is the only way a field's level is ever written: the sum over the sourcing families  $f$  of each family's own quantity  $q_f$  at the Node, weighted by  $w_f$ , plus the part's one remainder, divided by the part's wall  $E$ . The third line is what is read from a record and never written: its form  $D_i$ , which sources the fields; its Wronskian  $W_i$ , its charge; and its current  $F_{ij}$  into the Node  $i$  from its neighbour  $j$ , which a detector reads through its boundary in quanta of  $W_c = 3 \text{den } T$ , with  $T$  the quantum's norm, the normalisation of the quadratic form.

**The formulas derived.** From Eq. (1), in the paper's order, each with its mark in Table 1 and its algebra in the supplement: light's band and dispersion (Section 3.3). Light's index  $e^{2U}$  in a well (Section 3.4). The lattice Klein-Gordon equation and its Schrödinger limit with the inertia  $3 \tan \omega_0$  (Section 3.3). The conserved form and the share identity (Sections 4.3 and 4.4). The Wronskian as the charge (Section 5.5); Poisson's and Yukawa's rests,  $\cosh \kappa = 3 \text{den}/\text{num} - 2$  (Section 6.4). The bound body's functional and its stability (Section 7.2). The adiabatic invariant  $A^2 \sin \omega$  and the count's drift (Section 7.4). The clock's rate  $f(\omega_0) U$ , the bending  $4U$ , Shapiro's delay, the perihelion and the shadow's capture radius  $2e m$  (Section 8). The moving clock, Lorentz's factor reached as the family's kinetic scale reaches light's (Section 8, (e)). De Broglie's and Planck's relations and  $E = mc^2$  as  $m^* c_m^2 = \omega_0$  (Sections 8, (h), and 3.3). The content's index for every family, light's  $e^{2U}$  and matter's growing as  $1/k^2$ , the lens and the fall as refraction (Section 8, (p)); Bell's  $E$  and  $S = 478/169$ , Tsirelson's bound and GHZ's  $M = -4$  under the declared credit (Section 9.4). The fine-structure constant as a coefficient of the file,  $(3\sqrt{3}/8\pi) k/(\Gamma E_s)$  between two quanta with the sign holder's declared weight  $k$ , as  $G$  is read by the level weight; the hierarchy of the two forces between families with no number; the units and the highest photon energy (Section 10.1). And Newton's  $G$  as a

reading (Section 10.1). Fifteen formulas, several with their companions, from five fields, every one an algebraic consequence of the line and of the inputs named beside it.

**What the paper contributes.** We state the rule and prove what is exact about it, reversibility to the bit, the conserved form with its current, the momentum flux as the wave's own stress (Section 4). We formulate measurement as the meeting of the forward and the backward record at a declared detector, the one non-local element named (Section 5). We derive the families as bands of the rule and choose a universe with as few free numbers as the rule allows (Section 6). We show that bodies are fixed points, prove their stability under the dilation and state how they cluster (Section 7). We derive the weak-field forms of gravity and the moving clock's factor, each marked by what it rests on, in the model's own potential and in the one nature's tests read (Section 8). And we state the two slits and Bell as derivations of what each must give, neither experiment performed here (Section 9). The comparison with nature is partial by design. The moving clock, the weak-field forms, light's index, the two slits, Bell, the fine-structure constant and the units are compared at the order stated; the strong and the weak force, the lattice's frame and the nuclear binding are outside the present scope and are listed as open in Section 10.3; spin one half is reached as a route, its moment an explicitly named hypothesis. The method does not depend on the outcome of any one comparison: a family is one row of a file whose constants a reader may set and compute, and the implementation builds any universe of this kind from one file. This paper studies a single one, the rule's own, the matter pair  $[2, 3]$  over one denominator  $\Gamma = 6,000$ , four families, every divisor 1 but gravity's and the file's few numbers of nature and not of the rule, and reports where it agrees with nature and where it does not; what is general is the structure, a GameBoard of integers read into the physical world through clicks alone.

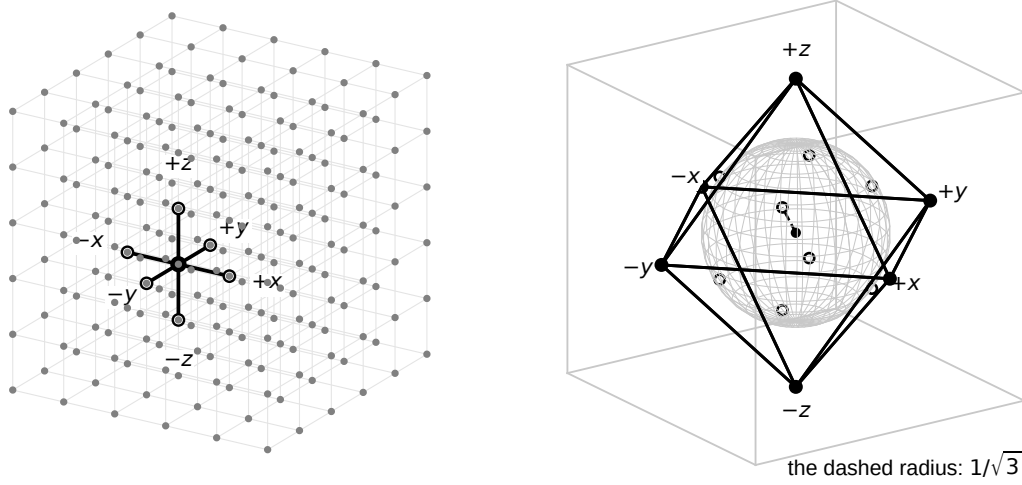
**What is put in, and the three kinds of claim.** The paper's claims are of three kinds, and the text keeps them apart. *Found*: what is proved or derived from the rule with no input chosen for it, the fifteen formulas above and the click formulas of Table 2 marked derived; the findings against nature are comparisons, listed in Section 10.3. *Assumed, chosen or declared*: the assumption and the credit below, the universe's rows, the detector's four declarations, its unit of credit among them, the electron as the free quantum of matter and the matter pair's denominator as the family's own integer (Section 10.1). *Held to be computable and not yet computed*: the bodies at the closing gap, the moving fixed point of a body and the absorbed fraction with the body's write; the explicitly named hypotheses are stated where they stand, among them the vacuum's write with its one declared number (Section 7.5), the three predictions entered as derived hypotheses (Section 10.2), the carrier that would mend the lattice's frame and the nuclear holders (Section 10.3). One assumption is named with the forms, that a Link is clocked twice by its reading Node's own clock,  $hN = 1$ ; the clock's composition and the acts' factors are derived from the rule, and under the sentence the model's static metric is the exponential metric, so the bending, the delay and the perihelion are Einstein's in the clock's own potential, the sentence itself fixed by those two clicks (Section 3.4); the forms against Kepler's potential reach Einstein's only as the matter family's gap closes (Section 8). One choice was made with a known result in view, and we state it as an input, not a prediction: a click's share is the square of the meeting, a credit chosen to reach nature's correlations, so Bell's  $2\sqrt{2}$  follows from it and does not test it. The file's numbers are nature's and not the rule's: the vacuum content, a declaration; the vacuum write's rate,  $\Lambda$ , the one declared number of the hypothesis of Section 7.5; the binding range, a world's choice; gravity's weight  $E_g$  and the sign holder's write weight  $k$ , read from  $G$  and from  $\alpha$ ; and the gaps, read from the masses (Section 10.2, Table 5). The magnetic force between moving charges is derived under the holder's rotation from the rule's own continuity identity, the sign's levels one covector, no number matched (Section 8). Table 1 lists every result of the paper with its kind, its fence and its inputs. What is not claimed: that nature is a lattice; that any number of the computed pair  $[2, 3]$  is nature's; that a comparison with nature is more than a consistency statement at the stated order; or that a formula known in the continuum is new where the paper gives its lattice form, the known form standing beside it in every row. Every derivation was checked twice, independently, within the project and, where a world exists, by the implementation's checks, not by an independent referee. Where the text says exact it means exact in the rational arithmetic before the rounding, with the integer remainders bounded as stated. The identifications of Section 10.1 rest on the electron as the free quantum of matter and change with it. And a line decided after a derivation is stated as the law with the implementation's run pending, a run that differed from its prediction being a defect of the implementation to repair and never a result.

**The three layers of the method.** The method has three layers, each with a fixed part and a free part. The board's rules (Section 3): which families and of what dimension, which holders, who reads what and how; Rule3 with its reads is fixed, the file's rows are free. The numbers (Section 6):  $\Gamma$ , the pairs, the holders' weights, gravity's  $E_g$  and the sign holder's  $k$ ,  $T$ , the vacuum content, the ranges, all free and as few as the rule allows. The click (Sections 5, 5.3 and 7.4): how the board becomes a click; what the detector declares, its window, region, floor and unit of credit; what is credited, the share, a bilinear form of the two records and so the double angle of Bell's  $E$ ; how a board mass is read as a click mass,  $M \sin \omega_b$ , the three masses; and the draw by the shares. Fixed here, common to every world,

**Table 1** The results of the paper, each with its kind, its fence and what it rests on. A calibration is a number taken from nature; EXPERIMENT a number the model does not fix.

| Result                                                                                                                                                                                                                                                        | Kind                                                         | Fence             | Rests on                                                                  | Section   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------|---------------------------------------------------------------------------|-----------|
| Rule3's step is reversible to the bit; the interval under the condition of Section 4.1                                                                                                                                                                        | theorem                                                      | GameBoard         | Rule3                                                                     | 4.1       |
| The gate says MATCH on the two slits', Bell's and GHZ's worlds: $N = 278$ , $S = 478/169$ , $M = -4$                                                                                                                                                          | computed, a run                                              | GameBoard         | the implementation                                                        | 4.1       |
| The rule's acts form no product of levels                                                                                                                                                                                                                     | theorem                                                      | GameBoard         | Rule3                                                                     | 4.2       |
| A conserved form, its current, and the momentum flux as the wave's stress                                                                                                                                                                                     | theorem                                                      | GameBoard         | Rule3                                                                     | 4.3       |
| The count is the record's share; the total share is never negative                                                                                                                                                                                            | theorem                                                      | GameBoard         | Rule3                                                                     | 4.4       |
| Three exact massive rotations                                                                                                                                                                                                                                 | theorem                                                      | GameBoard         | Rule3                                                                     | 6.1       |
| Light's speed $1/\sqrt{3}$ , its dispersion, and no dispersion along the cube's diagonal                                                                                                                                                                      | derived                                                      | GameBoard         | Rule3 for the pair $[1, 1]$                                               | 3.3       |
| A click is a meeting, credited by the shares; no collapse; no signal                                                                                                                                                                                          | assumption, derived                                          | clicks            | the meeting and the detector                                              | 5         |
| A field's reach from its gap, Yukawa's form                                                                                                                                                                                                                   | derived                                                      | GameBoard         | Rule3 for a gapped pair                                                   | 6.4       |
| The stability of a body under the dilation, to first order in the well                                                                                                                                                                                        | theorem                                                      | GameBoard         | the body's functional                                                     | 7.2       |
| Free fall is the same for extended bodies of one family to 2 to 8 percent in the computed regime                                                                                                                                                              | derived, computed                                            | GameBoard         | the fixed point under the massless field                                  | 7.3       |
| A body's clicks: no light from a steady rotation, light at a breathing body's beat, the resonance at $2\omega_b$ , the count's drift                                                                                                                          | derived                                                      | clicks            | Rule3 at the body's Nodes                                                 | 7.4       |
| A cluster: the three levels at a Node as the nesting, three kinds of binding, the click the only thing between clusters, the hierarchy recursive                                                                                                              | assumption, derived                                          | GameBoard, clicks | the universe is bound; Rule3                                              | 7.5       |
| The gravitational redshift, $z = f(\omega_0) U$                                                                                                                                                                                                               | derived                                                      | clicks            | the clock's read                                                          | 8         |
| The bending $4U_b$ , the delay, the perihelion's $6\pi U$ , $\beta = \gamma = 1$ ; the shadow's capture radius $2e m$                                                                                                                                         | derived under the one assumption                             | clicks            | a Link clocked twice by its Node's clock, $hN = 1$                        | 8         |
| The same forms against Kepler's potential: $\gamma_K = \text{den}/\text{num}$ , $\beta_K = (\text{den} + \text{num})/(2\text{num})$ , $1 + \alpha_K = \tan \omega_0/\omega_0$ , closing with the gap                                                          | derived                                                      | clicks            | the fall's formula                                                        | 8         |
| The moving clock: its fourth order and its kinetic scale $c_m$ , Lorentz's as the gap closes                                                                                                                                                                  | derived                                                      | clicks            | Rule3's band                                                              | 8         |
| The force between moving charges, Coulomb's over $\gamma$                                                                                                                                                                                                     | derived; the implementation's run a check of its bookkeeping | clicks            | the rule's continuity identity, the sign's covector                       | 8         |
| De Broglie, the fall, Newton's pull; the content's index for every family, the lens                                                                                                                                                                           | derived                                                      | GameBoard, clicks | the band and the field's rest                                             | 8         |
| Newton's pull, the clock's $G$ and Kepler's $G$                                                                                                                                                                                                               | derived                                                      | GameBoard, clicks | Rule3 for $[1, 1]$ and $E_g$                                              | 10.1      |
| $\alpha_{1\text{aw}} = (3\sqrt{3}/8\pi) k/(\Gamma E_s)$ between two quanta, $k$ the sign holder's declared weight; charge universality; the two forces' ratio between families $(m_p/m_e)^2$ ; the units; the highest photon energy; the gap below $10^{-18}$ | derived; $k$ declared                                        | clicks            | the rotation, Poisson's rest, de Broglie; the quantum of a family; LHAASO | 10.1      |
| The two-slit fringe row and the lattice's arrival interval                                                                                                                                                                                                    | computed                                                     | clicks            | Rule3 on the declared world                                               | 9.2       |
| Bell's $E = \cos 2(a - b)$ , $S = 2\sqrt{2}$ , $S(\rho)$ and Tsirelson's $2\sqrt{1 + \rho^2}$ ; every local credit at most 2 (theorem)                                                                                                                        | fitted; derived                                              | clicks            | the credit rule, the meeting's square                                     | 9.4       |
| The vacuum content 0; gravity's events at light's speed in every level                                                                                                                                                                                        | assumption; implemented                                      | clicks            | declared; every row reads the content                                     | 6.5       |
| The binding range, 20 Links                                                                                                                                                                                                                                   | a world's choice                                             | GameBoard         | the cloud of 1,836, a reading the families replace                        | 7.3       |
| Gravity's weight $E_g$ and the sign holder's $k$ , the matter pair's gap, the Links' wall                                                                                                                                                                     | calibration, experiment                                      | clicks            | nature's measurements                                                     | 6.6, 10.2 |
| The two clocks shift alike; the findings against nature of Section 10.3                                                                                                                                                                                       | derived; comparisons                                         | clicks            | the principle; Rule3                                                      | 10.3      |

is the click's one form: the meeting, the bilinear credit, the draw by the shares and the postulate that only a click is a measurement; free, the world's, are the detector's four declarations, the unit it counts in among them. So a gap at the free part, the counter's unit or how spin would show in a click, is open to construction, and a gap of the one form,



**Fig. 1** (a) The GameBoard in space: a cubic array of Nodes, here  $6 \times 6 \times 6$ , each joined to its six neighbours by Links through the Ports  $\pm x, \pm y, \pm z$  (the faint lines); one Node and its six neighbours marked, the Nodes one interval away from it. A Node keeps each family's two levels and its remainders there, and nothing else. (b) The Nodes one interval away from a Node form the octahedron  $|x| + |y| + |z| \leq 1$ , whose six vertices are the six Ports; its symmetry group is the cube's,  $O_h$ , the 24 rotations and, with the reflections, the 48 signed permutations of the axes, the group under which Rule3 is covariant (Section 3.1); its inscribed sphere, of radius  $1/\sqrt{3}$ , touches the eight faces on the cube's diagonals, and that radius is light's speed in Links per interval (Section 3.3). Both drawn from the definitions; no run.

no spontaneous click and so no lifetime, is fixed. Every comparison with nature passes through this layer.

**How to read the paper.** Every claim carries one of seven marks. **THEOREM:** proved for every lattice. **DERIVED:** obtained from theorems by argument. **COMPUTED:** a number obtained from the rule's lines before any run. **ASSUMPTION:** a line held by decision and named where it is used. **HYPOTHESIS:** a line outside the model, under its own name. **EXPERIMENT:** a number the model does not fix, with the experiment named that would fix it; the paper performs none. **FITTED:** an input chosen with a known result in view. The tables' status columns use the same marks, with calibration (a number taken from nature), declaration (a world's own number) and implemented (checked by the implementation) beside them. Every formula carries, beside its mark, its fence. **GAMEBOARD:** a reading of the lattice's own numbers that no instrument reports, a diagnostic. **CLICKS:** a formula of what a detector reports, the share  $Q_R/W_c$  on the forward orbit (Section 5.2) or a ratio of two such, the only kind that nature can be set beside. The algebraic steps of every derivation that the text states without its proof are given in the supplementary material (Online Resource 1), Derivations S.1 to S.47 in the order of the sections, cited as (S.*n*) where used. The runs of the implementation reported in this paper are the checks' (Section 3.7), the prediction written first and the run agreeing with it to the rounding, the implementation's check of the algebra and no experiment against nature. The law's document, the implementation, the derivations with their checks and the text were worked out with a large language model used as a tool under the author's direction, and the author reviewed every statement; the Statements and Declarations at the end say how. The paper cites three documents of its archive by the names of their sections: the law [1], the engine [2] and the decisions [3].

**The words.** The physical location is a *Node*, and its complete local information is its *NodeState*. Nodes are joined by *Links* through six directional *Ports* per Node, and the lattice of Nodes is the *GameBoard* (Fig. 1(a)). A local change is an *Event*, and there is one local rule, Rule3. A *family* is one kind of physical value with its pair and what it holds. Its *record* is its levels over the GameBoard, two per Node with a remainder. A family of quanta has a *dimension*, 1 for one real line and 2 for a plane of two lines. A *body* is the set of Nodes on which a family's record stands bound, with no law of its own; a *cluster* is bodies standing in one another's wells and clicking at one another. A *message* is a travelling record laid by the world, such as an emitter's packet of light. A *detector* is a region of Nodes declared in the file to report, and its *click* is one quantum credited to it from the current it saw through its boundary; the click is the one measurement. An *interval* is one step of the whole GameBoard. A family that holds a value for others to read is a *field family*, or *holder* [1, the objects][2, the words].

## 2 Related work

**Reversible cellular automata and digital physics.** The idea that the physical world is a discrete computation goes back to Zuse [4]; it was developed as reversible cellular automata by Fredkin, Toffoli and Margolus [5][6][7], as a programme by Wolfram [8], and as a deterministic foundation beneath quantum theory by 't Hooft [9][10]. Quantum cellular automata put a unitary step on a lattice, from the lattice Weyl, Dirac and Maxwell equations [11] and quantum lattice gases [12] to the structure theorems of the reviews [13][14] and the derivation of Weyl's and Dirac's equations from an automaton's symmetries [15]; the present rule's step is an integer recurrence on two levels, not a unitary on amplitudes. Rule3 is a second-order cellular automaton in the sense of Toffoli and Margolus,  $a_{\text{next}} = f(a_{\text{now}}) - a_{\text{before}}$  [7], with an integer alphabet and one Euclidean division whose remainder stays at the Node. Feynman noted that a local classical computer reproduces no correlations beyond Bell's bound [16]; the model agrees, and places the excess in the declared credit of the click rather than in the lattice (Section 9.4).

**Time-symmetric accounts of measurement.** Wheeler and Feynman described radiation as an interaction between the source and the absorber, with advanced and retarded fields together [17]; Cramer's transactional interpretation has an offer wave forward and a confirmation wave backward, meeting at the absorber [18]. The two-state vector formalism evolves one state forward from the preparation and one backward from the outcome [19][20]; Price argued that a time-symmetric account is what Bell's theorem leaves open [21], and Wharton and Argaman showed that every locally mediated reformulation of quantum theory must let some mediating parameters depend on measurement settings in their future [22]. The model belongs to this class, the backward reading from a realised click depending on the detector's setting alone; it differs in that both directions are one integer line, exact to the bit, that their product is a reading and not a wave, and that the choice among the meetings is the one non-local element; the record is an ontic state in the terms of [23].

**Bell's theorem and local models.** Bell's inequality bounds every local account of a pair's correlations [24], in the form that experiments test [25], and a local model, a joint distribution and the inequalities are one condition [26]. The model's local credits give  $S = \sqrt{2}$  and  $S = 2$ , as they must (Section 9.4); its credit by the contrast reaches the quantum correlation only by discarding pairs, Pearle's road [27], below the efficiency bounds of Eberhard and of Garg and Mermin [28][29], which the loophole-free tests of 2015 close for nature [30][31][32].

**Discrete gravity and its tests.** Regge calculus [33] and causal sets [34] discretise the geometry of space-time itself; here the lattice is never curved, and the weak-field forms are read from the coefficients (Section 8). In the parametrised post-Newtonian framework [35] the one assumption gives  $\gamma = \beta = 1$  exactly in the model's own potential, with the matter band's factors against the potential the tests read closing with its gap. A lattice breaks Lorentz invariance at order one unless its terms are tuned [36], and the bounds are reviewed in [37]; the paper's answer is the closing gap of Section 8.

## 3 The model

This section states the model: the lattice, the families, the rule, the paces through which one family acts on another, the one write, the readings and the order of acts in an interval [1, the GameBoard and the families; Rule3; the primitives].

### 3.1 The GameBoard

The GameBoard is a box of  $N_x \times N_y \times N_z$  Nodes. Two Nodes that differ by one step along one axis are joined by a Link, and each Node has six Links, one through each of its six Ports,  $\pm x$ ,  $\pm y$  and  $\pm z$ . Each axis is declared periodic, open or folded. Beyond an open face no Node lies and a level there is read as 0; a folded axis has one layer, and its two Ports return the Node itself. An inner face is a plane of Nodes declared closed except for declared gaps, which is how a wall with slits is made. A face may also be declared receding (ASSUMPTION): the box then grows by a layer of zeros whenever a nonzero level reaches the Node next to the face, so no event ever meets the face, and a run on such a box is a run on the unbounded lattice. The box grows and nothing stretches, so a receding face gives no cosmological redshift [1, the objects; the unbounded board][3, the receding face].

**The group and its action.** The symmetry group of the cube has 48 elements, the 24 rotations and, with the reflections, the signed permutations of the three axes,  $O_h$ , the octahedron's group as it is the cube's (Fig. 1(b)); with the translations it is the space group of the simple cubic lattice. It acts on the state space by permuting the Nodes and their Ports and carrying every level pair with its Node; the time reversal exchanges a record's present and its past and carries the step to its inverse (Theorem 1); and a rational rotation of a plane record's two parts,  $(p, q)/r$  with  $p^2 + q^2 = r^2$ , carries the rational solutions of the line without its division to solutions and leaves the Wronskian

unchanged. Rule3 commutes with the space group's action, since its line reads the six Ports symmetrically and names no axis: a world turned by an element of the group runs as the turned world. Under the group the form  $D$  is a scalar, even under the time reversal; the current  $F$  a vector, odd under it; the Wronskian  $W$  a pseudoscalar; and the click a scalar, since it names a region and an interval and never a Node or an axis. No boost, no rotation beyond the cube's 24 and no scaling is in the group, so Lorentz's group and the continuum's rotations are no symmetries of the lattice, and what the paper finds of the moving clock it finds as a property of the clicks (Section 8) (THEOREM for the covariance and the readings' kinds; GAMEBOARD; S.4) [1, the direction; the four acts][3, the one algebra].

The NodeState holds the model's own numbers and nothing else: for every family and every part, two levels and one remainder, and for every part of a field family, the remainder of its one write. A count, a current, a charge or a tension is a reading computed from these numbers when it is needed; no register, table, formula or random draw is kept at a Node [1, the objects][3, nothing at a Node but the law's numbers].

### 3.2 Families and records

A family is declared by two keys: its pair [num, den] and its shape. A family of quanta, such as matter or light, declares its dimension, 1 for one real line or 2 for a plane of two lines, and holds nothing. A field family declares its sources, the form, the form with the three tensions, or the Wronskian, which fix its parts, one real line per source, with its divisor  $E_s$  and, for the massless field, its rest; a field sourced by the form or the tensions holds the content, which slows the clocks of the families that read it, and a field sourced by the Wronskian holds the sign, the charge's field. A family's record is its levels over the GameBoard, at every Node a level now and a level one interval before, with a remainder; a family of dimension 2 lays two level pairs (re, im), two solutions of the same line, and only a plane carries a charge (Section 5.5). A body is a connected set of Nodes on which a family's record stands bound, and a message is a travelling record laid by the world [1, the objects].

### 3.3 Rule3

At every Node, in every interval, every family steps by the first line of Eq. (1). The six arrivals through the Ports, each weighted by its Link's coefficient  $R_{ij}$ , the Node's own level times  $S$ , minus  $w$  times the level before, plus the remainder the Node kept, are summed as one whole number and divided once by the wall  $w$ ; the quotient is the next level, and the remainder,  $0 \leq r' < w$ , stays at the Node and never travels. Nothing is rounded away, and the step back is the same division, which is why every run is exact and reversible to the bit (Theorem 1). The name comes from the three axes, one coefficient per direction serving its two Ports, the wall's factor 6 the three directions twice; for light the next level is the sum of the six arrivals over 3, minus the level before [1, the line].

$$a_{\text{next}} + a_{\text{before}} = \frac{\text{num}}{3 \text{ den}} \sum_a (\text{arr}_{+a} + \text{arr}_{-a}), \quad (2)$$

whose rotation at the wave vector  $\mathbf{k}$  is  $\cos \omega = \frac{\text{num}}{3 \text{ den}} \sum_a \cos k_a$ . At wave number zero  $\cos \omega_0 = \text{num}/\text{den}$ , and for the massless pair [1, 1] along one axis  $\cos \omega = (2 + \cos k)/3$ , so  $\omega = |k|/\sqrt{3}$  as  $k \rightarrow 0$  (DERIVED, GAMEBOARD; S.1). Light's speed at long wavelength is therefore  $1/\sqrt{3}$  Links per interval, below the causal bound of one Link per interval, and its group speed falls at shorter wavelengths as  $1 - 0.083 k^2$  [1, the lattice constants]. In nature's names the vacuum's line is the discrete Klein-Gordon equation, exact but for the remainder term: with  $S_6 = 6 + \Delta_{\text{lat}}$ ,  $a_{\text{next}} - 2a_{\text{now}} + a_{\text{before}} = c_K^2 \Delta_{\text{lat}} a_{\text{now}} - m^2 a_{\text{now}} + (r - r')/w$ , with  $c_K^2 = \text{num}/(3 \text{ den})$  and  $m^2 = 2(1 - \text{num}/\text{den})$ , the rest rotation as the mass ( $c_K$  is the equation's coefficient. The band's kinetic scale  $c_m^2$  of Section 8, (e), is  $\omega_0/(3 \tan \omega_0)$  and its largest group speed  $c \cos \omega_0$ , the three one as the gap closes). Its slow limit at small wave number is Schrödinger's equation with the inertia  $m^* = 3 \tan \omega_0$  and, in a content  $U$ , the potential  $\omega_0 f(\omega_0) U$  of Section 7.4 (DERIVED, GAMEBOARD). And  $m^* c_m^2 = \omega_0$  exactly for every pair: the model's  $E = mc^2$  at the band's kinetic scale, read in three click numbers, nature's its small-gap limit, the departure  $1 - \omega_0^2/3$  (DERIVED, CLICKS; S.18, S.21). At the vacuum's pace Rule3 is the leapfrog scheme of the Klein-Gordon equation, whose dispersion is the known one [38]; what the law adds is one rational [num, den] fixing mass and speed together, the equation exact, its departures predictions with bounds (Section 10.2), the integers with the remainder, and the paces (DERIVED).

### 3.4 The paces: how a field acts on a record

A field acts on a family by changing the family's paces at a Node, the coefficients of Rule3 there:

$$p_0 = \Gamma (1 - 1/\Gamma)^c, \quad p_i = \frac{p_0 (c_i)^2}{\Gamma}, \quad p_a(i, j) = \frac{p_i q_{ij}}{\Gamma}, \quad q_{ij} = \Gamma - t_a(i, j), \quad (3)$$

$$c = \sum_{\text{reads}} \text{weight} \times \text{the read family's time level},$$

Here  $c_i$  is the content the family reads at the Node, the sum over its reads of the weight times the read field's level. The clock  $p_0$  composes, each unit of the level slowing it by the same factor  $1 - 1/\Gamma$ , to the unit. The Node's pace  $p_i$  is the clock twice,  $p_0^2/\Gamma$ , so that the ruler goes with the clock;  $t_a(i, j)$  is the Link's tension, one number per Link from its two ends' parts (Section 4.3), read into the Link's factor  $q_{ij}$  the same from both ends. And the Link's pace is the product,  $\Gamma - 2c_i - t_a$  to first order [1, every row reads the content; the clock is the Node's, the tension is the Link's]. A positive weight makes a hollow, an attraction; a negative weight makes a hill. The pace is the clock's slowing: a record at a Node of pace  $p$  rotates and moves as at  $\Gamma$ , with its intervals  $p/\Gamma$  as long. Every family reads the holders of the content its declaration names at the weights it names, gravity's and the binding field's at the weight 1 for every family, the fields among the readers and a holder of the content its own level among it (the law's line, implemented; Section 3.7; the nuclear holders of Section 10.3 by the nucleon families alone), so  $x = (L_b + L_g)/\Gamma$  with  $L_b$  the binding field's level and  $L_g$  gravity's. A family of dimension 2 also reads the holder of the sign plainly, and light, the holder's own record, reads the holders of the content as every family does and not the holder of the sign. And a family sources what it reads at the weight it reads with, so every pair of families that read each other is reciprocal [1, the paces; the reads are the held row's pair's].

**The one assumption of the forms: a Link is clocked twice by its reading Node's own clock.** Nothing in Rule3 fixes how a level enters the paces; beside the line the model holds that a level is a count of units. Two parts of the paces follow from that. The clock composes,  $p_0 = \Gamma(1 - 1/\Gamma)^c$ , each unit slowing the clock as it stands by the same fraction, since the write of one unit is an act and every act is measured in the Node's own units, the read included (DERIVED). And every act of the interval, Rule3's step, the held rows' write and the sign holder's turn, is Rule3's act measured in the cell's units, so a uniform level  $U = c/\Gamma$  changes only the units, time by  $N = p_0/\Gamma$  and length by  $h = p_0/p_a$  per Link, the write carrying  $p_x p_y p_z / (p_0 \Gamma^2)$  and the turn  $p_0/\Gamma$ , per proper volume and per proper interval (DERIVED, S.34; implemented, Section 3.7). The one sentence not in Rule3, the model's one assumption of the forms, named as the credit is (Section 5): a Link is clocked twice by its reading Node's own clock,  $p_a = p_0^2/\Gamma$ , that is  $hN = 1$ , the ruler going with the clock (ASSUMPTION). The lattice says why twice, a Link having two ends, and not why the reading Node's clock stands at both rather than the two ends' product, which gives  $\beta = -1$ ; the sentence is fixed by two clicks, the bending's  $\gamma = 1$  and the perihelion's  $\beta = 1$ , and this is where Einstein's  $\gamma = \beta = 1$  enter the model by hand. Under it the model's static metric for light is  $ds^2 = -e^{-2U} dt^2 + e^{2U} d\mathbf{x}^2$  to  $O(U/\Gamma)$  and in the ray limit, the exponential metric [39], Einstein's to the first post-Newtonian order; the clock's rate is  $N$  in the lattice's variable  $2 \sin(\omega/2)$  exactly and in  $\omega$  to the finite-rotation correction  $\omega_0^2/12$ , 5.9 percent at the computed pair (6.3 exactly) and  $10^{-19}$  at nature's LHAASO bound ( $10^{-28}$  at the axis bound), the metric stated in the variable in which it is exact (DERIVED; S.31); the shadow's capture radius  $2e m$  its second-order click against Schwarzschild's  $3\sqrt{3} m$  (Section 8), in the lattice's frame, the frame a second residue not derived from Rule3 and read by the binary pulsars'  $\alpha_1$  (Section 10.3). The quadratic  $(\Gamma - c)^2 + c^2$  is the composition's second-order truncation [1, the same clock and the same ruler for every act; the paces compose][1, what is open, item 15].

**The band at a pace.** From the coefficients, the rotation of a record at a Node of clock pace  $p_0$  and Link paces  $p_a$  is (DERIVED, GAMEBOARD; S.1)

$$\cos \omega = 1 - \left(1 - \frac{\text{num}}{\text{den}}\right) \left(\frac{p_0}{\Gamma}\right)^2 - \frac{\text{num}}{3 \text{den}} \sum_a \left(\frac{p_a}{\Gamma}\right)^2 (1 - \cos k_a), \quad (4)$$

The mass term scales with the clock's square and the band term with the Link's. Two readings follow. First, at  $k = 0$  every massive band's rest rotation obeys  $1 - \cos \omega_b = (p_0/\Gamma)^2 (1 - \cos \omega_0)$ , the clocks' shifts of Section 7.4. Second, along a Link every wave's speed falls with the Link's pace,  $\Gamma - 2c - t_a$  to first order, so light is delayed and bent by twice the clock's share. At a Node of content  $x = L/\Gamma$  light's band is  $[2 - 4N^4, 2]$  in  $2 \cos \omega$  with  $N = (1 - 1/\Gamma)^L$ , and at long wavelength  $\omega = N^2 |k|/\sqrt{3}$ : light's speed is  $N^2 = e^{-2x}$  times its vacuum speed, to the unit, an index  $n = e^{2x}$ ,  $1/(1 - 2x)$  to the first order, with  $n - 1 = 2x$  (S.2). A record whose rotation lies outside the band at a Node is evanescent there and reflected whole; light meets a total mirror where  $p_a < \Gamma \sin(k/2)$  [1, the band at a pace; the rows against nature, (b)].

**The guard.** A pace is positive by the composition and must lie below an edge  $P$ , with  $P^2 = 2 \text{den} \Gamma^2 \text{div} (\text{den} + |\text{num}|)$ , which is  $\Gamma^2$  for light: beyond  $P$  the mode at wave number  $\pi$  on every axis grows without bound, and at or below  $P$  no mode grows exponentially (THEOREM, GAMEBOARD). The guard is the Courant, Friedrichs and Lewy condition of the explicit scheme Rule3 is at the vacuum's pace [38], checked once at load. Under the composed paces no Link ever closes: the clock  $\Gamma(1 - 1/\Gamma)^c$  is positive at every content and the Node's pace is its square over  $\Gamma$ , so a deep well freezes a clock and closes nothing, and the quadratic form's horizon at  $\Gamma/2$  is a feature of the truncation and not of the law (DERIVED, GAMEBOARD; S.3). A Node whose clock is frozen to the unit is cut from its six neighbours: no current crosses its Links and its share is not a number, so it is dark. The guard's bound,  $|2 \cos \omega| \leq 2$  at the band's

edges, admits the edge itself, where a linearly growing solution exists, and it is stated for  $\text{num} \geq 0$ ; at that edge the vacuum's checkerboard mode, driven by the remainders, grows as  $t^{3/2}$ , no click reading it, and reaches the loader's bound near  $1.4 \times 10^5$  intervals at  $\Gamma = 6,000$  (COMPUTED, GAMEBOARD), a negative numerator's lowest rotation lying at  $k = 0$  (S.3). At a negative content light's zone-corner mode grows, so light refuses a content below 0, and the vacuum's content is 0 (Section 6.5) [1, the guard; the frozen Node has no share].

### 3.5 The write, the readings and the interval

**The one write.** Every write into a field's level is one act, the second line of Eq. (1): at a Node in each interval a held part gains  $(\sum_f w_f q_f + r) \text{div } E$ , one division per part with one remainder  $r$  carried at that Node. Here  $q_f$  is the sourcing family's own quantity at the Node: its form  $D_f$  for a holder of the content, its Wronskian  $W_f$  for the holder of the sign, or its tension  $T_{aa,f}$  for an axis part.  $w_f$  is the weight with which that family reads the written field, and  $E$  is the part's wall,  $E_s T$  for the time part and  $E_s W_c$  for an axis part. The sources are summed in the numerator and never divided apart, so no remainder stands per sourcing family [1, the primitives][3, the one algebra].

**The readings.** A reading is a bilinear form of the levels with declared coefficients; it writes nothing on the GameBoard. The law reads three, the third line of Eq. (1). The form  $D_i = \text{now}_i^2 - \text{next}_i \text{before}_i$ , the present against the product of the future and the past, is  $B^2 \sin^2 \omega_b$  at every interval for a standing record of peak  $B$ ; its quotient over  $T$ , the well, sources the fields. The Wronskian  $W_i$  is the charge (Section 5.5). The current  $F_{ij} = \text{num}(\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$  into the Node  $i$  from its neighbour  $j$  is antisymmetric and, with the Link's factor  $q_{ij}^2/\Gamma^2$ , is the Link term of the conserved form (Theorem 3), the plain current where no tension stands; a wave travelling from  $i$  to  $j$  has  $F_{ij} < 0$ . A static record has  $\text{now} = \text{next} = \text{before}$ , so its form and its current vanish: a static field carries no quanta and no detector sees it [1, the booking; the conserved form].

**The interval.** Every read is of the values at the interval's start; Rule3's line reads one Link per interval, and nothing in zero time; every reading of the interval, the step, the tensions and the hold's write, is read from the record as it stands at the interval's start, so a Node's NodeState after one interval depends on the Node and its six neighbours alone, the interval's dependency radius one Link (the law's line, GAMEBOARD) [1, the interval's dependency radius is one Link]. The interval has four acts in order: (i) the read, each family's paces from the field levels, the reading field's own among them, and each Link's factor from its tension; (ii) Rule3 on every record; (iii) the readings, each record's currents, tension, form and Wronskian at the levels the step started from, and each detector's report; (iv) the write of each field from its summed sources. [1, the interval][2, the interval].

**What is conserved, and where.** On the GameBoard the conservations are per family: the weighted form of a family while its reads are static (Theorem 3), the Wronskian of a plane family under a static level, exactly in the line and to first order in a level's change (Section 5.5), the quanta through the Ports, the wave number of a free record and reversibility bit for bit. Energy is conserved within a family by theorem, between a charged body and its light by the law's line (Section 10.1), and between a body and the holders of the content it is replaced by the mass defect (Section 10.3). The GameBoard conserves no total: the sum of the families' forms is not conserved and not assumed, the share moving with the paces with no current to compensate (Eq. (13)). The click layer has books of its own: a click credits one whole quantum at the detector's own whole, drawn by the inflow of the record's share through its front over its window; a frequency of nature is read as the inflow's beat within the window or as the absorber's resonance; the GameBoard's quantities, the form, the content, the level, the rotation, are read by no click except as the share's inflow; and the credit's bookkeeping is that the shares sum to what the screen saw, one whole per click, the whole the detector's own transition (Sections 5.3 and 7.4). In the clicks there are conservations other than the board's, and neither layer keeps a total energy (THEOREM for the board's, ASSUMPTION for the credit's) [1, the conserved form].

### 3.6 The postulates

The model rests on seven postulates [1, the postulates]: (1) the world is Nodes and Events, an Event a level stepped by Rule3 with its remainder, a click a detector's report, and nothing else happens; (2) a Node holds bounded local information, read from itself and its six neighbours, and nothing of the past is kept; (3) consistency is local and causal, one Link per interval for every reading of the interval (Section 4.3), with one exception, the credit of a click by the shares over every declared detector, the one non-local element, not derived; (4) the causal speed is one Link per interval; (5) every physical calculation is on bounded integers, with no floating-point number, no root and no random draw in the law; (6) a measurement is a detector's click, the net current into a declared region through its front boundary over its window, never a Node; (7) the GameBoard, where no one measures, is deterministic and reversible to the bit, and the detectors' clicks are the only thing claimed to represent nature. The line and its clock are fixed:

every mend of the model is a family, a holder, a declaration or a read, never a change of Rule3 (ASSUMPTION) [3, nothing that is not Rule3 remains].

### 3.7 The implementation

The implementation, Universe24 [40][2], is the law's implementation and nothing else: a GameBoard of Nodes, every family's NodeState at every Node, and one interval of acts in the law's order, each a call of Rule3 on whole-board arrays of integers at a declared width, every neighbour read through a Port. It holds no number of physics, no formula, no family's name and no flag, every physical value coming from the run's files; a run is headless, writes the click lines, the measurement, and runs for results only after the check of Section 4.1 says MATCH [2, the engine; the NodeState; the output; the gates]. It holds the law's lines of Section 3.4, the composed paces, the Link's factor, every row at its reads' paces, the write's and the turn's factors and the guard's two sides; the checked worlds' outputs under them are bit-identical to the quadratic forms' runs [2, the NodeState; the interval]. One prediction of the law is stated for a three-dimensional run not yet performed, the one-dimensional chain not carrying the ruler: a body in a uniform level, a well of at least 300 levels at  $U = 0.05$ , its rest line's and its beat's  $2 \sin(\omega/2)$  ratio to  $N$  within 1.5 over the well's depth in levels, an absolute bound [1, what is open, item 29]. The checks' runs: the two slits, the Huygens prediction 273, written before the run, the run  $N = 278$ , and the law's real line 277.8 (278.0 in the physical labels), computed after the run by the script `paper/general_formula/two_slits_real_line.py`; Bell,  $S = 478/169$  with the marginals  $1/2$ ; GHZ, the eight shares, every pairwise  $E$  0 and  $M = -4$ ; and the back-in-time check MATCH on every world (COMPUTED, runs of the implementation; CLICKS) [1, the rows against nature, (g) and (h); the GHZ gate]. The paper reports the checks' runs and derives the rest: a run of a declared world computes what the algebra of that world states, the implementation held to the law's lines by its tests and the gate.

## 4 Exact properties

This section proves what holds exactly on every lattice: the reversibility of the rule, the kind of function its acts can form, the conserved form with its current and its momentum flux, and the count as a reading of the record.

### 4.1 Reversibility

Let  $\sigma$  be the direction,  $+1$  forward and  $-1$  backward; let  $\text{div}$  and  $\text{mod}$  be the floor division and its remainder in  $[0, w)$ ; and let  $\rho$  be the remainder carried in. With

$$u = \sigma \left( \sum_j R_{ij} \text{arr}_j + Sx - wy \right) + \rho, \quad z = \sigma(u \text{ div } w), \quad \rho' = u \text{ mod } w, \quad (5)$$

the forward call with  $(x, y, \rho) = (a_{\text{now}}, a_{\text{before}}, r)$  gives  $(z, \rho') = (a_{\text{next}}, r')$ , and the backward call with  $(x, y, \rho) = (a_{\text{now}}, a_{\text{next}}, r')$  gives  $(z, \rho') = (a_{\text{before}}, r)$ .

**Theorem 1** (the direction). *The backward call returns  $(a_{\text{before}}, r)$  bit for bit; the reversibility of every step needs no second function.*

*Proof.* Forward,  $\sigma = 1$ , the line is the definition of  $\text{div}$  and  $\text{mod}$ . Backward,  $\sigma = -1$ : write  $Y$  for the right side's first two terms,  $\sum_j R_{ij} \text{arr}_j + Sa_{\text{now}}$ . Then  $u = w a_{\text{next}} - Y + r'$  and  $z = -\lfloor u/w \rfloor = \lceil (Y - w a_{\text{next}} - r')/w \rceil$ . From the forward line,  $Y - w a_{\text{next}} - r' = w a_{\text{before}} - r$  with  $0 \leq r < w$ , whose ceiling divided by  $w$  is  $a_{\text{before}}$  exactly; and  $\rho' = u - w \lfloor u/w \rfloor = w a_{\text{before}} - (Y - w a_{\text{next}} - r') = r$ . [1, the direction].  $\square$

The whole run therefore goes back:  $N$  intervals forward and  $N$  back return every array of every family bit for bit, provided every number an act read at an interval's start is one the state after the interval still holds (THEOREM, by construction from Theorem 1). The implementation checks this as a test, running a world forward and back and comparing every array at every interval; on the worlds of Section 9 it reports MATCH over their whole runs, worlds whose write stays under one level (Section 6.6); the interval's reversal with a level-changing write is not yet gated (COMPUTED, GAMEBOARD) [1, the inverse; the objects][2, how to run a world][3, the Node knows no family]. The condition rules out one read: a coefficient read from a Node's own state across a step cannot be recovered at the step back, so a record's rotation sense is never read at a Node and the holder of the sign is read by the record's shape, a plane or a line (Section 5.5) [1, the interval][3, the board goes back in time, as a gate; there is no sense].

### 4.2 What the acts can form

Rule3's line is one act, and declared coefficients make it four: (a) the read, the line with its six arrivals; (b) the one-Node step, the line with the arrivals declared 0, which rotates a level by Chebyshev's recurrence; (c) the division,

the line with  $R = S = 0$ , Euclid's division with the remainder kept; and (d) the load, a declared integer in the numerator, which is how a count enters a level [1, the four acts].

**Theorem 2** (what the acts can form). *Every composition of the four acts is a piecewise-linear function of the levels with integer slopes: a sum of linear forms and of floors of linear forms, composed. No such function is quadratic in the levels.*

*Proof.* Each act is a linear form of the levels or the floor of one; a floor of a linear form is piecewise linear with integer slopes; sums and compositions of such functions are of the same kind. The second difference of such a function in any one level is 0 almost everywhere, while the second difference of a product  $\text{now}_i \text{before}_j$  in the pair  $(\text{now}_i, \text{before}_j)$  is 1. So nothing bilinear comes from the acts alone.  $\square$

So every product of two levels is a reading and never a write, and the three readings of Section 3.5 are the only products the law takes. A line enters the law only if it passes three tests: it is generic (one primitive with declared integers, no family's name), vector (an act on the whole state, with no root and no floating-point number) and local (its own record and the six neighbours, nothing kept at a Node beyond the law's numbers). A line that needs more is a hypothesis under its own name. No act of the interval takes a square root; the integer square root appears only when a world is laid, as the fixed point of the division  $\text{act } x \leftarrow (x + n \text{ div } x) \text{ div } 2$  [1, the three tests of every rule][3, the six verbs; the root leaves everywhere].

### 4.3 The conserved form and the momentum flux

The read coefficient  $R_a(i \rightarrow j) = 2 \text{ num } p_i^2 q_{ij}^2 / \Gamma^2$  carries the Node's factor and the Link's; the Node's factor is not symmetric where the paces differ and the Link's is, so the conserved form carries the weight  $1/p_i^2$  on its Node terms and the Link's factor  $q_{ij}^2 / \Gamma^2$  on its Link terms:

$$E = \sum_i \frac{w (\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i}{p_i^2} - 2 \text{ num } \sum_i \sum_{j \sim i} \frac{q_{ij}^2}{\Gamma^2} \text{now}_i \text{before}_j. \quad (6)$$

**Theorem 3** (the conserved form). *At constant paces for which a positive Node weight  $d_i$  with  $d_i R_a(i \rightarrow j) = d_j R_a(j \rightarrow i)$  on every Link exists,  $E$  with the weights  $d_i$  is exact under the line without the rounding. Under the law's read coefficient, the Node's pace squared times the Link's factor squared,  $R_a(i \rightarrow j) = 2 \text{ num } p_i^2 q_{ij}^2 / \Gamma^2$  with  $q_{ij} = q_{ji}$  and  $p_i = p_0(c_i)^2 / \Gamma$  the composed Node factor, the weight  $d_i = 1/p_i^2$  cancels the Node's factor on both sides of every Link, so  $E$  is exact for any field and any tensions. With the rounding each Node's remainder adds at most one unit of the level per interval to the walk of  $E$ . Its Link term carries the Link's factor, so the current of Rule3's own conservation law through the Link  $ij$  is  $\text{num } (q_{ij}^2 / \Gamma^2) (\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$ , the plain current  $F_{ij}$  where no tension stands,  $q_{ij} = \Gamma$ ; the form with the plain current holds there alone, and with a tension it is not conserved (two witnesses, S.32).*

*Proof.* Without the rounding the line is  $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M} \mathbf{a}_{\text{now}}$  with  $\mathbf{M}$  the matrix whose diagonal is  $S_i/w$  and whose entries on the six Links are  $R/w$ . Let  $\mathbf{D}$  be the diagonal matrix of the weights  $1/p_i^2$ ; then  $\mathbf{D}\mathbf{M}$  is symmetric iff  $d_i M_{ij} = d_j M_{ji}$  on every Link,  $d_i R_a(i \rightarrow j) = d_j R_a(j \rightarrow i)$ : with  $R_a(i \rightarrow j) = 2 \text{ num } p_i^2 q_{ij}^2 / \Gamma^2$  the weight  $1/p_i^2$  cancels the Node's factor on both sides of every Link and the Link's factor  $q_{ij}^2$  is the same from both ends, so  $\mathbf{D}\mathbf{M}$  is symmetric for any field and any tensions. For a symmetric  $\mathbf{K} = \mathbf{D}\mathbf{M}$  the quantity  $Q(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot \mathbf{D}\mathbf{x} + \mathbf{y} \cdot \mathbf{D}\mathbf{y} - \mathbf{x} \cdot \mathbf{K}\mathbf{y}$  satisfies  $Q(\mathbf{a}_{\text{next}}, \mathbf{a}_{\text{now}}) = Q(\mathbf{a}_{\text{now}}, \mathbf{a}_{\text{before}})$ : with  $\mathbf{a}_{\text{next}} = \mathbf{M}\mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}}$ ,  $\mathbf{a}_{\text{next}} \cdot \mathbf{D}\mathbf{a}_{\text{next}} - \mathbf{a}_{\text{next}} \cdot \mathbf{K}\mathbf{a}_{\text{now}} = \mathbf{a}_{\text{before}} \cdot \mathbf{D}\mathbf{a}_{\text{before}} - \mathbf{a}_{\text{before}} \cdot \mathbf{D}\mathbf{M}\mathbf{a}_{\text{now}}$ , which is the other side. Multiplied by  $w$  this is  $E$  up to the Link term's sign convention [1, the conserved form].  $\square$

$E$  is a diagnostic. Where the paces move, the total form  $D = |\mathbf{a}_{\text{now}}|^2 - \langle \mathbf{a}_{\text{next}}, \mathbf{a}_{\text{before}} \rangle$  changes exactly by  $D(t+1) - D(t) = \langle \mathbf{a}_{\text{next}}, (\mathbf{M}(t) - \mathbf{M}(t+1)) \mathbf{a}_{\text{now}} \rangle$ , which is 0 at fixed uniform paces and negative where a well deepens under a standing record (THEOREM, by substitution; GAMEBOARD) [1, the conserved form].

**The momentum flux is the wave's own stress.** The momentum density along the axis  $a$  at the Node  $i$  is the current per axis,  $P_a(i) = (F_{i,i+a} - F_{i,i-a})/\text{num}$ , and the line without its division gives exactly, at constant paces, the identity below; the integer step adds the remainder term  $\delta_i \Delta_a(\text{now})_i - \text{now}_i \Delta_a(\delta)_i$  with  $\delta = r - r'$  and  $\Delta_a$  the centred difference, which the implementation's test includes (THEOREM for the line, GAMEBOARD),

$$w [P_a(t+1) - P_a(t)] = \sum_b R_b [G_{ab}(i-b) - G_{ab}(i)], \quad (7)$$

$$G_{ab}(i) = \text{now}_i (\text{now}_{i+a+b} - \text{now}_{i-a+b}) - \text{now}_{i+b} (\text{now}_{i+a} - \text{now}_{i-a}),$$

with  $G_{ab}(i)$  the stress reading on the  $b$ -Link  $(i, i + b)$ , the levels now alone. The proof substitutes the line into  $w P_a(t + 1)$ : the  $S$  term cancels, the  $w$  term is  $w P_a(t)$  and the  $R$  term is the difference of the fluxes. The tension on the axis  $a$  at a Node, the part a field reads into the Link's pace and the source of a field's axis part, is the momentum flux averaged over the Node's two  $a$ -Links,

$$\begin{aligned} T_{aa}(i) &= -\text{num} (G_{aa}(i) + G_{aa}(i - a)) \text{ div } 2, \\ G_{aa}(i) &= \text{now}_i (\text{now}_{i+2a} - \text{now}_i) - \text{now}_{i+a} (\text{now}_{i+a} - \text{now}_{i-a}), \end{aligned} \quad (8)$$

and each Node sources its own part,  $-\text{num } h_a(i)$  with  $h_a(i) = \text{now}_{i-a} \text{now}_{i+a} - \text{now}_i^2$ , read from its two neighbours along the axis at the interval's start; the Link's tension is its two ends' parts, one number per Link read the same from both ends by the pace at the next interval. A plane wave of amplitude  $b$  at the wave number  $k$  has  $G_{ab} = -2b^2 \sin k_a \sin k_b$ , so the flux along the wave is positive, as a wave's pressure is (S.7); a packet of count  $c$  and velocity  $\mathbf{v}$  has  $c v_a v_b$ , the stress of dust. With the sign as written a wave's stress deepens the content around it, so light gravitates by its pressure too. The off-diagonal parts and the vector part, which Rule3 never reads, are not written [1, the tension; the final unification][3, the tensor part is the record's own stress].

#### 4.4 The count is the record's share

A family's count of quanta is a reading of its record and no line of its own. At every Node the record's share is

$$e_i = \left[ w (\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i - \text{now}_i \sum_{j \sim i} R_{ij} \text{before}_j \right] \text{div} (2p_i^2), \quad (9)$$

the conserved form's term at the Node with its weights (Theorem 3), the Node's  $1/p_i^2$  and each Link's  $q_{ij}^2/\Gamma^2$  inside  $R_{ij}$ , read in quanta over  $W_c$  as the nearest whole number; at the vacuum's paces it is  $3 \text{den} (\text{now}_i^2 + \text{before}_i^2) - \text{num now}_i S_6(\text{before})_i$  exactly,  $S_6$  the six-neighbour sum, and read with the vacuum's coefficients instead of the Node's own paces the same expression is the vacuum's share, the unit of the tables of Section 7. The implementation reads it so; the law's document prints the Node term over the mean of the Link paces squared with the plain Link term, the same where no tension stands. Over one step of Rule3 the share's rational value, before its floor, changes by exactly the six currents into the Node, each weighted by its Link's factor, plus Rule3's own rounding (THEOREM for the rational identity, GAMEBOARD; S.5); the integer share of Eq. (9) adds the reader's own floor, a residual strictly between  $-1$  and  $1$  per Node and level pair, distinct from Rule3's remainder:

$$e_i(\text{next}, \text{now}) - e_i(\text{now}, \text{before}) = \sum_{j \sim i} \frac{q_{ij}^2}{\Gamma^2} F_{ij} - \frac{(\text{next}_i - \text{before}_i)(r'_i - r_i)}{2p_i^2}, \quad (10)$$

with  $F_{ij}$  at the levels the step started from and  $r, r'$  Rule3's remainder before and after the step. So over any region the total share changes only by the weighted currents through the region's boundary, the remainder terms and the readers' floors (THEOREM, GAMEBOARD) [1, the count is the record's share; the share's change is the currents]. A formulation moving whole quanta by a division of the current would put numbers at every Node that no act of Rule3 reads back; the law does not carry it (ASSUMPTION) [3, the count's line and the sense's line leave the law]. Where the paces move the share drifts with no current crossing any Link, Eq. (13), a diagnostic [1, the paces' own change moves the share with no current].

**The signed integers.** The levels are signed integers, the currents are signed by their direction and the Wronskian by its sense, and a click is one whole quantum, never negative. The total share of a closed or periodic GameBoard is non-negative for  $|\text{num}| \leq \text{den}$ , with equality for a static uniform level of light and, on an even periodic board, for its checkerboard (THEOREM, GAMEBOARD; S.6). One Node's share can be negative at a lay's edge, and a region's by at most half a level-square per boundary Port. Over the whole passage of one forward packet the inflow accumulated at a region is non-negative up to the rounding, since the lay's current is forward at every Link, so the credit by the shares is well defined; where a window is cut inside a passage the instrument credits  $\max(s_R, 0)$  (DERIVED, CLICKS) [3, the law's integers are signed].

## 5 Measurement: the click as a meeting

The click is the meeting of the future with the past at a declared detector, and no act of the GameBoard. This section states what is on the lattice, the meeting, the detector, what a click leaves behind, the charge and light, and what locality costs [1, the click; the reading].

## 5.1 Events, and no waves

On the GameBoard there are events and amplitudes. An event is a level split by Rule3 to the six neighbours; an amplitude is the level Rule3 produces at a Node, the sum of the split events there; and the record is the amplitudes over the GameBoard. In our world there are clicks, the detectors' reports, and nothing else. In the two slits the emitter's events pass both gaps, every path of the future at once; the fringes are where split events from the two gaps meet at the screen with the same sign. The past is made of the same split events: read backward from the realised click by the same line, Rule3 at  $-1$ , they form the past's tree, which also passes both gaps back to the emitter. What stays on the GameBoard, the amplitudes no click realised, is not in our world (ASSUMPTION) [1, no waves on the event board][3, no waves on the event board].

## 5.2 The click is the meeting of the future with the past

**Assumption 1** (the click is a meeting, the law and not a hypothesis). A record runs on the GameBoard in both directions by the one line: forward from the emitter, the possibilities, the future; backward from the realised click, what happened, the past, Rule3 at  $-1$ . A click is where the two carry the same signal at a declared detector; its share is the product of the two directions at the region, the form  $D = \text{now}^2 - \text{next} \times \text{before}$ , the now against the future times the past; which meeting is realised is the draw, the one non-local place. The detector only asks whether the future and the past met at it.

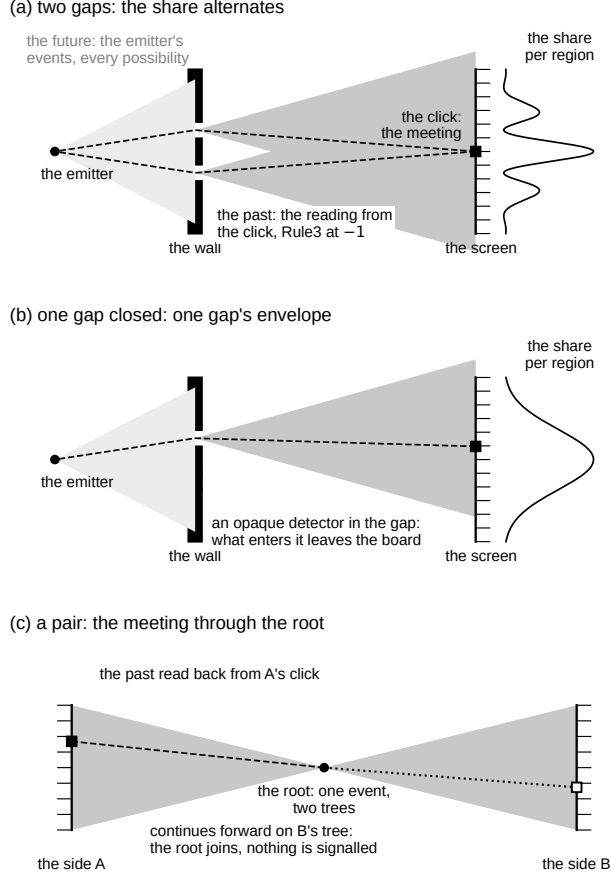
Computationally, by the linearity of the line, the backward reading from a realised click equals a sum over the forward records, so the meeting adds a reading of what a click is and no mechanism of choice; which meeting is realised stays the draw by the shares, the one non-local place.

In plain words (Fig. 2): a click happens when the past meets the future; the click is the now; the detector only asks whether the future and the past met at it; and a click is created only at their meeting. This is a line of the model and not a hypothesis [3, the click is the meeting of the future with the past].

**The click, algebraically.** Let  $X$  be the state space: at every Node, for every part of every family, one integer level pair and the write remainders, and nothing else. The step  $S_t$  is Rule3 over the interval with the paces read from the fields at its start, a bijection of  $X$ , exact by the division with its remainder; the line without its division is linear in the levels at fixed paces, the linearity the backward reading uses, the integer map not affine; its forward orbit is the future and its inverse orbit the past. The readings  $D$ ,  $F$  and  $W$  are bilinear forms on  $X$ , covariant under the symmetry group of Section 3.1 and never written. An instrument  $R$  is a declared region with its front Ports and its coefficients, which define  $Q_R$ : for one line a quadratic form of the levels, the inflow through its front Ports accumulated over the window; for a pair, the window's sum of the squared cross-side sum over the parts, quadratic in the joint amplitude, the product of the two sides' levels, and so quartic in the levels. A click is the pair  $(R, t)$ , the credit of one whole quantum to the instrument  $R$  at the interval  $t$ , drawn with the probability  $s_R / \sum_{R'} s_{R'}$ , where  $s_R = Q_R(x_t) / W_c$  is the instrument's form on the forward orbit over the quantum's measure of a current. In the meeting's words, the inverse orbit from the credited state coincides with the forward orbit on the realised path, and the share is the bilinear form's value at the coincidence. The click writes nothing to  $X$ : it is the one map in the model that is not a bijection, from the vector of shares to a unit vector, and it is a scalar under the group, so no click names a Node. Every observable is  $Q_R / W_c$  on the forward orbit or a ratio of two such. The window is the whole passage of a packet, or the credit floors at 0 (Section 4.4); a Node whose pace rounds to 0, a frozen Node, stands in no instrument's front and is never clicked (ASSUMPTION for the definition, DERIVED for the precisions; CLICKS) [3, the click's algebraic definition].

**One form for every click.** At the credit the parts of a record are read with the detector's coefficients, each part paired with the same part through the common origin, summed and squared; the share is that density, and the draw is by the shares. The two slits are the case with nothing to pair: light is one real line, and the share is the record's own inflow. Bell's check is the case through the common origin: a pair born as one event is two trees with one root, and the backward reading from the first click passes the root and continues forward on the partner's tree (Section 9.3). In the implementation the joint share is computed from the forward records alone, by the linearity of the rule, the check of Section 4.1 showing that the backward reading exists to the bit (DERIVED, CLICKS).

**No duality.** What is called the particle as a wave is the future: the record run forward by Rule3 with every possibility, which has not yet met the past. What is called the particle as a particle is the present: the click, one whole quantum credited at one region in one interval. The particle's track is the past: the chain of realised clicks, and the same record read backward through them. There are not two things but one record read in two directions, and the point where the two readings coincide is what appears as a particle. After a click no wave vanishes and nothing



**Fig. 2** The click as the meeting of the future with the past, schematically (ASSUMPTION; drawn from the definitions, no run). (a) The two slits: the emitter's events spread forward through both gaps, every possibility at once (the grey fans, the future); from a realised click at a region of the screen the same line is read backward, Rule3 at  $-1$ , through both gaps to the emitter (the dashed lines, the past); the share along the screen, the product of the two directions, alternates. (b) One gap closed by an opaque detector: its events leave the declared board; the backward reading meets one gap and the share is one gap's envelope. (c) A pair, two trees with one root: the backward reading from A's click passes the root and continues forward on B's tree (the dotted line); the root joins and nothing is signalled.

collapses: the record goes on, the click is a credit, and the past has grown by one click (ASSUMPTION) [3, the wave is the future, the click is the present, the track is the past].

### 5.3 The detector is a declared instrument

**Assumption 2** (the detector's declaration). A detector is one connected region of Nodes declared in the file, never one Node, at least half the wavelength of the family it reports across the beam (4 Links at  $\lambda = 8$ ); a narrower region, a single Node or a region in pieces is refused at load. It is opaque, backed by a receding face or by the wall's own face, so that what enters it leaves the declared board and never returns. Its click is its report, per interval and family: the net current into the region through its front boundary Ports, the Ports leading in from the declared board, summed in integers, with the region's name and the family and never a Node; one wall  $W_c = 3 \text{ den } T$  of the inflow over a passage is one quantum, so a screen saw  $N$  quanta where its regions' inflows sum to  $N W_c$ . The credit: one quantum to one detector, drawn by the shares of the regions' inflows, read outside the GameBoard over every declared detector of the world, one credit per quantum and never two. The credit is the one non-local place of the description and no line of the law.

Why a region: a record confined to one Node contains every wave number at once,  $\Delta x \Delta k \geq 1/2$  on the lattice as in nature; so no click names a position finer than a region, and every per-Node number is a diagnostic [1, the click is the detector's report; no click names a Node][3, no click names a Node].

What produces a click is deterministic: the meeting, the current of the amplitudes through a region's boundary. What is not known is which of the possibilities becomes the click: when one possibility is spread over several detectors,

the GameBoard gives the shares, no line of the law gives which detector fires, and the instrument draws. Exclusivity, one click per quantum, and Born's rule, the credit's chance equal to the intensity's share [41], are therefore declarations of the instrument and not findings of the model [1, what produces a click is not known]. The uncertainty relation is not a separate rule: it is the detector's three declarations, a region of at least half a wavelength, a credit per interval and one draw per quantum (ASSUMPTION) [3, the uncertainty relation is not a rule of its own]. What each declaration costs: a region of  $n$  Nodes reads a fringe of period  $\Lambda$  with the factor  $\sin(\pi n/\Lambda)/(n \sin(\pi/\Lambda))$ ; a frequency is read to  $\Delta\omega \geq 2\pi/N$  over  $N$  intervals; and  $N$  quanta drawn by the shares scatter as  $\sqrt{N p_i(1-p_i)}$ , so a pattern is in the histogram and never in one click (DERIVED, CLICKS; S.8) [2, the output].

## 5.4 The click ends nothing

**Assumption 3** (the click ends nothing). The click is a detector's reading, local to its boundary; the record goes on after it and nothing ends anywhere. What a detector absorbed leaves the declared board through the face that backs it; the other possibilities go on splitting and leave through the receding faces, events that were not realised, in no click. The ending of a record at a click by the detector is decided against and is no hypothesis of the law.

This is the model's claim and not a deficiency: perhaps nature is local. No signal passes, since nothing at one detector changes the amplitudes at another. A measurement needs no collapse: the record never ends, the click is a reading, and the backward reading from the click writes nothing [3, the click ends nothing]. A bound body can act as an instrument in the same form: the future is light's forward record at its boundary, the past is the body raised by one quantum read backward to the source, and the meeting is the same frequency, the resonance of the body's rotation with the light's (DERIVED, CLICKS; Section 7.4). The body's write back waits on the fixed point with the write [3, a Node of the board is read only through a detector][1, the rows against nature, (y)].

## 5.5 The sign is the rotation sense

A family of quanta laid as a plane carries its record as two level pairs, (re, im), stepped by one call of Rule3 with one set of coefficients, each with its own remainder; a field's parts stay real. The charge density at a Node is the reading

$$W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}}, \quad (11)$$

the Wronskian of the two pairs, which the line without its division conserves exactly with the conserved form's weights, under Theorem 3's condition on the paces (THEOREM, since **DM** is then symmetric; GAMEBOARD; S.9). The integer step adds to each part the remainder term  $\epsilon = (r - r')/w$ ,  $|\epsilon| < 1$ , so the integer  $W$  changes at a Node by  $\epsilon_{\text{re}} \text{im}_{\text{now}} - \epsilon_{\text{im}} \text{re}_{\text{now}}$  per interval, not zero in general: at the vacuum's pace of [2, 3] the parts (1, 3) with zeros before step to (1, 4) and  $W$  from 0 to  $-1$  (COMPUTED, GAMEBOARD; S.9); an exactly conserved integer charge is open (Section 10.3). Its local change is the Link current,  $(2 \text{num}/w) (q_{ij}^2/\Gamma^2) (\text{im}_i \text{re}_j - \text{re}_i \text{im}_j)$  per Port for the weighted  $W_i/p_i^2$ ,  $\text{num}/(3 \text{den})$  times the bracket for the integer  $W_i$  at the vacuum's paces (S.9), and its total over the GameBoard is invariant under any change of the paces that keeps them uniform at each Node, since the bracket  $\mathbf{im}_{\text{now}}^\top \mathbf{M} \mathbf{re}_{\text{now}} - \mathbf{re}_{\text{now}}^\top \mathbf{M} \mathbf{im}_{\text{now}}$  vanishes for every symmetric **M** (THEOREM, GAMEBOARD). A real record ( $\text{im} = 0$ ) has  $W = 0$ , so light, the real record of the holder of the sign, is neutral; a rotating record has  $W$  of either sign by its sense [1, the sign is the rotation sense; the sign's count is the record's own].

Who reads the holder of the sign is decided by the family's dimension and by no name (ASSUMPTION, GAMEBOARD): a family of dimension 2 has a Wronskian and reads it, a family of dimension 1 has  $W = 0$  and does not. The holder of the sign does not read its own level, so light does not read it and a static charge does not bend light. A reader's own sense never enters its coefficients, since a coefficient read from a Node's own state would break the exact inverse (Section 4.1). So the holder of the sign acts on its readers by the act its row declares: **rotation**, the holder's level over  $\Gamma$  as the angle  $\theta$  turning the two-part record in its own plane,  $e^{-i\theta} z_{\text{next}} + e^{i\theta} z_{\text{before}} = (S z_{\text{now}} + \sum_a R_a (z_{+a} + z_{-a}))/w$ , three exact integer shears inverted bit for bit with  $\theta$  read at now alone, under which like and unlike senses are told apart by the record's own sense. Which sense moves apart is one bit the model does not fix, declared by nature's click that like charges repel, the shears' exact angle  $2 \arctan(L/2\Gamma)$ ,  $L/\Gamma$  at small angles, and the magnetic force follows as the Link's phase, Peierls' coupling (Section 8) (ASSUMPTION for the declared bit, GAMEBOARD) [3, there is no sense; every family has a dimension; the rotation round, built][1, the sign holder rotates the two-part record; the rows against nature, (f)].

## 5.6 Light is born by the write

Light is the holder of the sign's own record: one record whose time part is a real level pair, stepped by Rule3 at the paces of its reads, so that light is bent by the content, and written by the one write from the Wronskian of the records that read it. A static rotating body writes a static level, Coulomb's field, and radiates nothing. A breathing

or moving body writes a varying source at its Nodes, the field carries it away at one Link per interval, and those spreading events are the light (DERIVED, CLICKS). A detector reads light's current through its boundary in quanta of  $W_c$ , and the travelling events of the other fields, which in gravity's case nature calls a gravitational wave, enter a detector the same way. A body's quanta that leave it stay its family, and no quantum changes family at a crossing. A body reads the holder of the sign at its own Nodes, so a breathing body reads the light it writes and its pace answers, the radiation reaction, in the same loop [1, light is born by the write][3, light is born by the write].

## 5.7 The price of locality

Two consequences of the model's locality, stated against nature's clicks [1, what is open, item 4; the rows against nature, (g) and (h)][3, the click is the meeting of the future with the past].

- (i) *Exclusivity and Born's rule are declared.* A quantum spread over several detectors is a set of shares of the current at each; the model gives the shares and the instrument gives the click.
- (ii) *Bell.* A pair is one laid record, a common cause, its phase a shared classical variable inside Bell's hidden variable [24]; every local credit of that phase gives  $|S| \leq 2$  (THEOREM, Bell's applied to the model's credits; CLICKS; Section 9.4) [25], and the meeting through the pair's root gives  $E = \cos 2(a - b)$  and  $S = 2\sqrt{2}$  with nothing signalled, resting on the credit rule, a declaration (ASSUMPTION, FITTED; CLICKS). [3, the click is the meeting of the future with the past].

A which-way measurement without absorption is no third price: nature has no detector that marks a path without a click, the model's opaque detector in one gap loses the fringes as nature does, and a bare region's inflow is a GameBoard diagnostic and never a measurement [3, go over the paper and see which mismatches with nature the clicks resolve].

## 5.8 The formulas of clicks

Table 2 collects the formulas of clicks the model gives, each with its status and the section that derives it [3, the click's algebraic definition; the mathematics of the bound bodies and how it bears on the clicks].

# 6 The families and the universe

A family is a band of the one rule and not a separate law. Its row in the universe file holds only what the rule and the geometry do not fix: its pair [num, den], with  $\cos \omega_0 = \text{num}/\text{den}$ , and its shape: for a family of quanta its dimension, for a field its sources with its divisor. A pair written over the GameBoard's one denominator  $\Gamma$  makes a mass one integer  $m$ , with  $\cos \omega_0 = m/\Gamma$  and the wall  $6\Gamma^3$ ; a reduction to lowest terms is a reading and never an act of the loader, so such families share one resolution of their remainders [1, the families from the rule; the mass is one integer; the wall is one].

**How a family enters.** A family enters the world as one row of the universe file with two keys and nothing else: its pair, and its shape, which for a family of quanta is its dimension, 1 or 2, and for a field its sources, the form, the form with the tensions, or the Wronskian, with the field's divisor  $E_s$  and, for the massless field, its rest. Its rotation at rest, its band, its coefficients and its wall,  $w = 6 \text{den } \Gamma^2$ , follow from Eqs. (1) and (4); its parts are its sources' count, one real line per source; its reach is set by its gap (Section 6.4); its reads are its declaration's, gravity's and the binding field's at the weight 1 (Section 3.4), and it sources what it reads at the weight it reads with. Every record clicks, a field's included, since no row declares whether it clicks. Table 3 gives the rows of the universe, and Section 6.6 the numbers of the world [1, a family's declaration; the families from the rule][2, how to add a family].

## 6.1 The exact bands, and the third band

The line of one Node at rest,  $a_{\text{next}} = 2 \cos \omega_0 a_{\text{now}} - a_{\text{before}}$ , is exact in integers only where  $2 \cos \omega_0$  is an integer. So the rule carries exactly three exact massive rotations,  $2 \cos \omega_0 = 1, 0$  and  $-1$ , with the periods 6, 4 and 3 and the pairs [1, 2], [0, 1] and  $[-1, 2]$  (THEOREM, GAMEBOARD). A record on an exact band at rest in the vacuum leaves no remainder in its rest rotation, so it never clicks and only turns. The exact bands are exact only in the vacuum: at a Node of pace  $p_0$  the band's rest rotation is  $2 \cos \omega = 2 - 2g(p_0/\Gamma)^2$  with  $g = (\text{den} - \text{num})/\text{den}$ , an integer at  $p_0 = \Gamma$  alone, so inside a well a record on an exact band leaves a remainder and clicks like every family's [1, the exact bands; the exact bands are exact only in the vacuum].

The bands [0, 1] and  $[-1, 2]$  are allowed by the rule and unused by the universe [1, the third and the colours, closed as absent].

**Table 2** The formulas of clicks, each with its status. Every row is a formula of what a detector reports, CLICKS.  $U$  is the clock's share and  $U_K$  the Newtonian potential the orbits measure (Section 8).

| The formula                        | Status                                                                     | What the model gives                                                                                                                                                                                                                                                           | Section  |
|------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Born's rule, the count at a region | derived                                                                    | $N_R = Q_R/W_c$ , the share of the inflow through the region's front over the window; the pattern is in the histogram, $Np_i \pm \sqrt{Np_i(1-p_i)}$                                                                                                                           | 5.2, 5.3 |
| The dead time                      | declaration                                                                | not derivable: the shares credit a region any whole number of quanta in one interval, and $W_c$ over the inflow's rate is the mean spacing of the clicks and not a least one                                                                                                   | 5.3      |
| The two-slit row                   | derived; computed                                                          | the near-field minima at the rows 15 and 33; the screen's inflow peaks at the interval 68.0 against the fourfold refined lattice's 61.8                                                                                                                                        | 9.2      |
| The arrival and its spread         | derived; computed                                                          | $t = L/v_g$ with $v_g = (\text{num}/(3 \text{ den})) \sin k / \sin \omega$ ; the click's duration $\sigma_t = \sqrt{\sigma_x^2 + (\omega'' L \sigma_k / v_g)^2} / v_g$ from the band's curvature, 4.13 intervals at $k = \pi/4$ over $L = 34$ Links; the centre's arrival 62.2 | 9.2      |
| Bell's pairs                       | the credit declared (Born's rule for a pair); $E$ and $S$ derived under it | $E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$ and $S = \sqrt{2}(1 + \rho)$ , 478/169 at the integer settings for equal parts                                                                                                                                                    | 9.4      |
| GHZ, three quanta                  | derived under the declared credit (S.40)                                   | $E_3 = \cos 2(a + b + c)$ , every pairwise $E$ 0, Mermin's $M = -4$ where every local credit gives $ M  \leq 2$                                                                                                                                                                | 9.4      |
| The clock's click rate             | derived                                                                    | $1 - f(\omega_0)U$ , $f = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$ , $1 - U$ at a small gap; the beat to $(1 - N^2)\omega^2/24$ ; a cavity of bodies the same, declared faces $N^2$                                                                                      | 7.4, 8   |
| The bending's clicks               | derived under the one assumption; in $U_K$                                 | $4U_b$ ; $(1 + \text{den}/\text{num}) 2GM_K/(c^2b)$ against Kepler's potential                                                                                                                                                                                                 | 8        |
| The absorption line                | derived (S.16, S.43); the one-mode line under the pace read alone          | parametric at $2\omega_b$ , the growth $\epsilon/(4 \sin \omega_b)$ ; the transfer line at $\omega_j - \omega_i$ , Rabi's rate, under both reads                                                                                                                               | 7.4      |
| Emission                           | derived                                                                    | a steady rotation and a uniform motion write no light; a breathing body writes at its beat, at the first order in the breathing's depth                                                                                                                                        | 7.4      |
| The which-way sum                  | derived                                                                    | the continuity of the one inflow through the two fronts: a region behind one gap reads that gap's share and nothing of the fringe                                                                                                                                              | 5.7      |
| The uncertainty relation           | declaration, with its inequalities                                         | $\Delta y k \geq \pi$ from the region, $\Delta \omega \geq 2\pi/N$ from the interval, $\sqrt{Np(1-p)}$ from the draw                                                                                                                                                           | 5.3      |
| No signalling                      | derived                                                                    | $P(A+) = (\cos^2 a + r^2 \sin^2 a)/(1 + r^2)$ , independent of $b$ for parts of any amplitudes                                                                                                                                                                                 | 9.4      |
| A body's count in clicks           | derived; the unit of credit declared                                       | the share summed over the body's Nodes, what a full absorber credits; under a moving well it drifts by Eq. (13); a body's click one transition, $\hbar\omega$ per click                                                                                                        | 7.4      |
| Magnetism                          | derived under the holder's rotation (S.33); not under the plain read       | $F_C/\gamma$ : the sign's halved current over the wall den $T$ ; 0.995 at 0.1c and 0.866 at 0.5c                                                                                                                                                                               | 8        |

## 6.2 The rule's own universe

The universe of the model is the one whose every number is the rule's own: the pairs from the integers 1, 2 and 3 of the three exact massive rotations and the three axes, every divisor 1 except gravity's, and the remaining free numbers named. Its rows stand in Table 3, as the file `examples/events/rule.json` holds them (DERIVED from the two keys of each row) [1, the families from the rule; the rule's own universe][2, the input].

The universe's integers are the Node clock  $\Gamma = 6,000$ , a multiple of 6 so that the pairs of light and matter are exact over it; the quantum's norm  $T = 2^{15}$ , the normalisation of the quadratic form; and the integers' width, 63 bits, a bound and not physics.  $\Gamma$  and  $T$  are units, as  $c$  and  $\hbar$  are in nature, and no line of Rule3 fixes  $T$ , since the rule is linear and the scale of its form is free. Matter reads gravity and the binding field plainly,  $x = (L_b + L_g)/\Gamma$ ; charged matter, a second row of the matter pair at the dimension 2, reads the holder of the sign as well; light reads the same holders of the content and is bent by them, and does not read the holder of the sign, its own field. The bound charge's field, the exact band [1, 2] at under one Link's reach, was read by neither of the implementation's two worlds and has left the universe file (ASSUMPTION) [3, nothing that is not Rule3 remains]. Every field of the file is a record stepped by Rule3 at its pair and written by the one write; no field is laid by hand during a run, and a world's starting rest is the fixed point of the division act, computed once before the first interval (ASSUMPTION) [1, the final

**Table 3** The rule’s own universe, its four families as the file `examples/events/rule.json` holds them (DERIVED from the two keys of a row, its pair and its shape): each family with its pair [num, den] (in brackets over  $\Gamma = 6,000$ ), what it holds at its divisor, its shape, what it reads, and its free number. A further family, the pair family of light, two real lines never summed, is declared by Bell’s world and not by this file (Section 9).

| Family                                                                                                              | Pair                           | Holds, at the divisor                                                                                                                 | Shape                                                                                               | Reads                                                                                          | Its free number                                              |
|---------------------------------------------------------------------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| gravity, the time part and the three tensions                                                                       | [1, 1]<br>([6000, 6000])       | the content, $E_g$ (1,000 in the bodies’ worlds; $10^6$ in the two slits’ and Bell’s worlds, gravity off in all but name) the sign, 1 | 1 + 3 real lines, sourced by the form and the three tensions                                        | the content, its own level among it (the law’s line, implemented)                              | $E_g$ , the hierarchy; its rest $c_{vac}$ , 0 by declaration |
| the charge, the holder of the sign, whose quanta are light, born by the write                                       | [1, 1]                         |                                                                                                                                       | 1 real line, sourced by the Wronskian of its readers                                                | the holders of the content, at the weight 1; never its own level                               | $k$ , the write weight, 1 in the file                        |
| the binding field, not the strong force (Section 10.3); its own events have the gap                                 | [2400, 2401]                   | the content, 1                                                                                                                        | 1 real line, sourced by the form                                                                    | the content, its own level among it (the law’s line, implemented)                              | the pair, its range $R = 20$ Links, a world’s choice         |
| $\cos \omega = \text{num}_b / \text{den}_b$ matter, the scalar; a quantum that leaves a body is a message of matter | [ $m$ , den]<br>([4000, 6000]) | nothing                                                                                                                               | dimension 1, one real line; a charged body of the same pair is a second row of dimension 2, a plane | the holders of the content, $x = (L_b + L_g) / \Gamma$ ; the plane also the holder of the sign | $m$ , the mass                                               |

unification][3, two units and the bounds; nothing that is not Rule3 remains].

### 6.3 What each family carries, and how gravity enters as a clock

What a family carries is set by what Rule3 reads and what a record can write, not by a name (DERIVED, GAMEBOARD). Rule3 reads two things of a Node and no third: the clock’s pace  $p_0$ , one number, and the Link’s paces, the three directions of the rule’s vector, so a field that acts on the paces carries at most one time part and three axis parts, 1 + 3. A record of one real line, neutral matter or light, has a form  $D$  and a stress and no Wronskian; a record of a plane, charged matter, has a Wronskian  $W$  as well; a field is always real, one line per source, since a plane’s own  $W$  would be its own source. The shapes follow with no choice: *the scalar*, one real line, for matter, light and the binding field; *the plane* for charged matter; *one plus three*, the time part with the three tensions, for gravity, sourced by the form and by the diagonal of the record’s own stress (Section 4.3); and *two real lines never summed at a Node* for the pair family of light in Bell’s world. The law as it stands takes no vector part of a field and no off-diagonal stress, since Rule3 reads them nowhere; for a real record no current’s read exists within the three tests, so no frame dragging of neutral bodies (Section 10.3) [1, the final unification; the tension][3, every family has a dimension].

**How gravity enters: a slowed clock, and no curved space.** There is no metric, no force and no law of motion in the model. Gravity enters in three steps. (i) *The write*: at every Node in each interval every family of quanta writes its form over the action,  $D_i \text{ div } T$ , into gravity’s field over the divisor  $E_g$ , by the second line of Eq. (1). (ii) *The field*: gravity’s field, [1, 1], steps by Rule3 at the pace 1, so around a static source it rests at Poisson’s solution,  $3G(r)s$ , which is  $3s/(4\pi r)$  far away; a change travels at one Link per interval, retarded. (iii) *The read*: every family reads the field’s level  $c$  at its Node into its own coefficients, Eq. (3), the clock’s pace once and each Link’s pace twice. What nature calls curved space is here the Link’s pace, light’s index  $n = e^{2x}$ ; no space is bent. The factor two on the Link is the one assumption of Section 3.4, the clock’s composition the rule’s own [1, the paces; the clock once and the Link twice][1, the same clock and the same ruler for every act; the paces compose].

### 6.4 The reach of a field, and the binding field

A field’s rest solves  $\text{num } \Delta^2 a - 6(\text{den} - \text{num}) a = -3 \text{den } \sigma$ , its line with the source once an interval, with  $\Delta^2$  the six-neighbour second difference. Outside a body its rest therefore falls by  $e^{-\kappa}$  per Link with  $\cosh \kappa = 3 \text{den} / \text{num} - 2$ , the family’s own band at zero rotation: the gap is both the mass of the field’s quanta and the inverse reach of the field,

one number (DERIVED, GAMEBOARD).  $[1, 1]$  has no gap, and its rest is Poisson's over the whole GameBoard.  $[1, 2]$  has  $\cosh \kappa = 4$  and falls by 0.127 per Link, a contact well of  $1/\kappa = 0.48$  Links' reach. A pair with num just under den has  $\kappa^2 = 6 \text{ den/num} - 6$  to second order, a range  $R = 1/\kappa = \sqrt{\text{num}/(6(\text{den} - \text{num}))}$  Links, which is Yukawa's form [43] (S.11)  $[1, \text{the well}]$ .

**Assumption 4** (the binding field and gravity's divisor, decided with a reservation). The well of a record,  $D_i \text{ div } T$ , is laid into two fields. The binding field, a family with a gapped pair  $[\text{num}_b, \text{den}_b]$ ,  $\text{num}_b < \text{den}_b$ , at the divisor 1, stepped by Rule3 at its pair as the massless field is and started at its rest, which with the gap is a sum with a range,  $R = \sqrt{\text{num}_b/(6(\text{den}_b - \text{num}_b))}$  Links and  $e^{-r/R}/r$  beyond it: it binds the bodies and is not the strong force (Section 10.3). Its own travelling events have the gap  $\cos \omega = \text{num}_b/\text{den}_b$ , a quantum with a mass whose Compton length is  $R$ , Yukawa's own line [43] and no input. The massless field  $[1, 1]$  at the divisor  $E_g$  declared by the file: gravity, long-ranged and weak by  $E_g$ . Every family reads the holders its declaration names, at the weights it names, and sources each at the weight it reads with: gravity's and the binding field's levels are read by every family at the weight 1, the two fields among the readers, each its own level among the content, and the nuclear holders of Section 10.3 by the nucleon families alone (the law's line, Section 3.4; the selective read a declaration, ASSUMPTION).

The first pair is  $[2400, 2401]$ , with  $R = 20$  Links, and the field's quantum has the rest frequency 0.029 per interval. The range is a world's choice: a cloud of 1,836 quanta is bound under  $R = 20$  and not under  $R = 10$  or 15 (Section 7.3), a reading of the proton the families replace (Section 10.3). In the implementation's worlds  $E_g$  is declared large, 1,000 in the bodies' worlds, so that gravity does not enter a body's binding; the falls of Table 4 were computed under the massless field alone at  $E_g = 1$ , on a world of an earlier round not among the shipped files, and stand as computed there. Nature's  $E_g$  is read from  $G$  as the sign holder's  $k$  is from  $\alpha$ , and the hierarchy of gravity and electricity is the product of the two declared weights and the quantum's energy against its unit of charge (Section 10.1). The decision carries a reservation, that it may be wrong [3, the two rows][1, what is open, item 20].

## 6.5 The vacuum content

**Assumption 5** (the vacuum content is 0, a declaration fixed by nature's clicks). The massless field rests at a declared level everywhere, the vacuum content, the key **rest** of the universe file, derived by no line of the law; declared 0 in the file's integers at  $\Gamma = 6,000$ .

A uniform rest  $c_{\text{vac}}$  is a change of units (Section 3.4), which only nature's clicks can tell. Under the law's line every field reads the content as every family does, so the massless field's own travelling events have light's paces identically in every level: gravity's events and light through the same wells arrive together, the difference of their two Shapiro delays 0 and  $\gamma_{\text{GW}} = \gamma_{\text{EM}}$ , where GW170817 bounds  $\gamma_{\text{GW}} - \gamma_{\text{EM}}$  between  $-2.6 \times 10^{-7}$  and  $1.2 \times 10^{-6}$  [44] (DERIVED from the law's line; CLICKS; S.12). Inside matter the source is read through the pace, gravity gravitating (DERIVED, GAMEBOARD). With the two speeds one speed the measurement bounds no rest, and the vacuum content stays a declaration, 0 in the file and, under the vacuum's write of Section 7.5, the faces' positive content  $\Lambda R^2/6$  (ASSUMPTION).  $[1, \text{every row reads the content; what is open, item 22}][3, \text{the vacuum content is 0, a declaration fixed by nature's clicks}]$ .

## 6.6 The world's numbers, and how the universe was chosen

The universe is chosen by one principle stated before any row: one rule closes everything; the families are its inputs, each a band of the rule named by its pair; the units are a choice of scale; and the universe holds as few free numbers as the rule allows (ASSUMPTION) [3, the standing goal; not one number in the engine, and as few in the files as the law allows]. The road to them:

(i) a family is a band of the rule, and its row holds only its two keys; (ii) three exact massive rotations, the pairs  $[1, 2]$ ,  $[0, 1]$  and  $[-1, 2]$  (THEOREM, Section 6.1); (iii) the three of Rule3 is the three axes, so the massless holder of the content carries 1 + 3 parts and every other family is a scalar; (iv) one denominator for light and matter, their pairs written over  $\Gamma$ , a multiple of 6, the binding field's pair the one not over it; (v) the pairs from 1, 2 and 3 (ASSUMPTION):  $[1, 1]$  for gravity and the charge's field, and  $[2, 3]$ , the first pair past the exact bands, for matter, nature's pair far past it (Sections 8 and 10.1); (vi) every divisor 1, under which the universe carried no free number but the mass; (vii) two numbers where nature's scale forced them, the binding field's gapped pair and gravity's divisor  $E_g$  (ASSUMPTION, with the reservation of Section 6.4); (viii) the vacuum content 0 where no write is declared (Section 6.5); (ix) the sign holder's write weight  $k$ , 1 in the file, read from nature's  $\alpha$  as  $E_g$  is from  $G$  (Section 10.1).

What remains: two units,  $\Gamma$  and  $T$ ; one bound, the integers' width; and five numbers of physics, matter's numerator  $m$ , the binding field's pair, gravity's weight  $E_g$ , the sign holder's write weight  $k$  and the vacuum content, and under the hypothesis of Section 7.5 one more, the vacuum write's rate, with every other divisor 1. The two worlds read a light-only copy of the file with the two holders of the content at the divisor  $10^6$ , so that light's own well stays under

one level. That this universe is the universe is a decision and not a theorem (ASSUMPTION); the electron's mass over Planck's and the proton's mass over the electron's are open and not derived (Section 10.3) [1, the rule's own universe; the final unification][3, the rule's own universe is the universe; the final unification].

## 7 Bodies

### 7.1 What a body is, and how it is computed

A body is its record over its Nodes and the quanta that record carries, and never one Node. Its count is the share of its record (Section 4.4), a reading and not a state; its form  $D_i \text{ div } T$  is written into the fields it sources, and the fields enter Rule3 through the paces alone. The record is bound by the wells its form sources, by the binding field's screened sum with the range  $R$ , while gravity's  $1/r$  is weak by  $E_g$  and enters no body's binding. No law is written for a body: it is the support of a standing mode of the one rule, and its record moves by Rule3 alone. Four statements make a body (DERIVED from the line, with the decisions named): (a) the body is the fixed point of its binding field, standing where the sources return themselves; (b) a field's level is the rest the start solves from the bodies' forms, after which Rule3 steps it; (c) a body's quanta are its record's share, carried wherever its record's current carries it; (d) a free packet or a cloud falls as  $a = \Delta |d\omega_b/dc| \nabla c$ , and a compact body by its tail alone, more slowly [1, what a body is; the four lines of the body][3, what a body is].

**How a body is computed.** The held rows' source is the record's form  $D_i \text{ div } T$ , the law's line; the implementation's start laid the count, so every well of this section was laid deeper than its hold returns by the factor  $e^{4U}$ , the hold returning  $e^{-4U} = 0.72$  of it at the compact body's centre, and its numbers stand under that start until recomputed (DERIVED; GAMEBOARD). By the rule's own iteration and no other formula: start from a spread of the declared count over the Nodes; write the wells that spread lays into the fields and let them rest; find the record's top mode in those paces by Rule3's own read act, iterated; scale its two levels so that the share carries the declared count; write the wells that record lays; and repeat until the levels repeat, in integers, at the fixed point of the division act. The branch is chosen by the start's width alone: a wide start reaches the cloud and a narrow one the compact body, two fixed points at one count. [1, the generator; the four lines of the body][3, the generator is Rule3].

### 7.2 Stability

Let a body of  $M$  quanta have the record  $\phi$  over its region,  $\sum \phi_i^2 = 1$ , with the counts  $c_i = M\phi_i^2$ . Let every field  $r$  it sources be a hollow ( $\sigma_r = +1$ ) or a hill ( $\sigma_r = -1$ ), with the static kernel  $\mathbf{G}_r$  of its line and the divisor  $E_r$ , so that the well at a Node is  $\Phi_i = \sum_r \sigma_r (M/E_r) \sum_j G_{r,ij} \phi_j^2$ . To first order in  $\Phi/\Gamma$  the well raises the one-Node rotation  $2 \cos \omega$  by  $4g\Phi/\Gamma$ ,  $g = (\text{den} - \text{num})/\text{den}$ , and the body's fixed point is a critical point on the sphere of

$$F[\phi] = \frac{\text{num}}{3 \text{ den}} \sum_{\text{Links}} (\phi_i - \phi_j)^2 - \frac{2g}{\Gamma} \sum_r \sigma_r \frac{M}{E_r} \sum_{ij} \phi_i^2 G_{r,ij} \phi_j^2, \quad (12)$$

the band's pressure less the wells' energy, each well counted once because the record is its own source (Hartree's form). The record is the top mode of Rule3's symmetric form at the paces, and  $dF/d\phi = 2\lambda\phi$  on the sphere is the fixed point's own equation with  $2 \cos \omega_b = 2 \cos \omega_0 - \lambda$  (THEOREM, GAMEBOARD; S.13). Let  $\phi_R(x) = R^{-d/2} \phi(x/R)$  be the body dilated, with  $d$  the GameBoard's dimension; the pressure is  $K/R^2$  and each field's well energy is  $W_r/R^{s_r}$ , with  $s_r$  the field's scaling power at the body's size:  $s_r = d$  beyond the field's reach and  $s_r = d - 2$  within it or for the massless field [1, the functional and the dilation].

**Theorem 4** (the stable body, under the dilation and to first order in the well). *At a fixed point of the binding row of width  $R$  the virial  $2K = \sum_r \sigma_r s_r W_r$  holds, and the second variation of  $F$  along the dilation is  $R^2 F''(R) = \sum_r \sigma_r s_r (2 - s_r) W_r$ . A body is stable only where this sum is positive; it is a minimum of  $F$  where further the Hessian on the modes orthogonal to the dilation and the translations is positive.*

*Proof.*  $F(R) = K/R^2 - \sum_r \sigma_r W_r R^{-s_r}$ ;  $F'(1) = 0$  is  $-2K + \sum_r \sigma_r s_r W_r = 0$ ;  $F''(1) = 6K - \sum_r \sigma_r s_r (s_r + 1) W_r = \sum_r \sigma_r (3s_r - s_r^2 - s_r) W_r$ : Derrick's scaling [45], Pohozaev's identity [46] on the GameBoard [1, the functional and the dilation].  $\square$

The theorem is a statement about the functional of Eq. (12), to the first order in the well, along the one direction of the dilation, with the dilation taken as continuous; it is not a theorem of the integer step on the full lattice, and the Hessian beyond the dilation is open (Section 10.3).

The factor  $s(2 - s)$  is  $+1$  at  $s = 1$ ,  $0$  at  $s = 2$  and  $-3$  at  $s = 3$ : a hollow holds the body where its well falls more slowly than the pressure and breaks it where it falls faster. Three consequences follow. (a) One hollow beyond its

**Table 4** The cloud, the wide branch of the binding row's fixed point, along its branch (DERIVED from the fixed point, a reading of the algebra and no click; GAMEBOARD): the total count, the rotation  $2 \cos \omega_b$  against the band's top 1.3333, the rms radius in Links, the count at the centre Node, the well and the inertia over the count, and the fall against a free packet's. An open 41-cube at  $\Gamma = 6,000$ , matter [4000, 6000], under the massless field alone at  $E_g = 1$ , a world of an earlier round not among the shipped files; the counts in the vacuum's share.

| Total count | $2 \cos \omega_b$ (top 1.3333) | rms radius | Centre Node | Well / count | Inertia / count | Inertia / well | Fall / free |
|-------------|--------------------------------|------------|-------------|--------------|-----------------|----------------|-------------|
| 4,000       | 1.3509                         | 8.7        | 2.6         | 0.985        | 1.061           | 1.077          | 0.92        |
| 3,000       | 1.3437                         | 10.3       | 1.0         | 0.991        | 1.037           | 1.047          | 0.95        |
| 2,000       | 1.3383                         | 11.5       | 0.4         | 0.995        | 1.022           | 1.027          | 0.97        |
| 1,500       | 1.3359                         | 12.0       | 0.3         | 0.996        | 1.016           | 1.020          | 0.98        |

reach binds every mass on a chain and in a box has one critical point, a saddle: the body disperses beyond it and collapses within it. (b) A massless field holds the body: Poisson's well in a box has  $s = 1$ , so  $F = a/R^2 - c/R$  has one critical point,  $R = 2a/c$ , a minimum for every mass, which is Pekar's body [47]. Beside it a hollow beyond its reach adds a barrier,  $F = a/R^2 - b/R^3 - c/R$  on a Gaussian body, with the wide critical point a minimum and the narrow one a saddle, and none where  $3bc > a^2$ , the collapse (COMPUTED, GAMEBOARD). (c) A collapse ends in a frozen clock: beyond first order the well saturates and the Link's pace weakens the pressure, and since no pace reaches 0 before the rounding a collapse ends in a cluster of Nodes whose clocks are frozen to the unit and never in a closed Link [1, the functional and the dilation][1, what is open, item 29].

### 7.3 The two branches: the cloud and the compact pixel

The fixed point of the binding field has two branches at every count above an edge, and the body of nature is the wide one (DERIVED from the fixed point and computed, GAMEBOARD; the identification is a decision of the model) [1, the two branches][3, the body of nature is the cloud].

**The cloud, the body of nature.** The wide branch is the continuum's Pekar body [47]: the count is spread over many Nodes, with the Pekar radius  $5.25 \Gamma E_g / C$  Links for a count  $C$  in the continuum's scaling, within a tenth of the table's for the two heaviest counts, the box limiting the lighter two. Its three masses agree within a few percent: the differences go as  $(C/\Gamma)^2$  for a cloud free of the box and linearly for a cloud the box limits (COMPUTED, GAMEBOARD). Table 4 gives the three along the branch (an open 41-cube,  $\Gamma = 6,000$ , the counts in the vacuum's share) [1, the rows against nature, (u)].

So the universality of free fall [35] holds in the model for extended bodies to 2 to 8 percent in the computed regime, quadratically in  $C/\Gamma$  for a free cloud and linearly for a box-limited one, and the better the larger  $\Gamma$  is against the body's count.

**The electron and the proton.** The electron is one quantum of the matter family laid with a sense, a message and not a body, stable because one quantum does not disperse into fractions. The proton is computed here as a cloud of 1,836 quanta, the reading Section 10.3 replaces by one quantum of its own family, so  $m_p/m_e = 1,836$  is a ratio of counts and fixes no  $\Gamma$ ; the model does not derive it, and the proton's count is an input, declared in the world file (ASSUMPTION). A body's mass is a click mass: its inertia is read by its click period and de Broglie's  $k$ , its weight by the fall's clicks, its well by another body's orbit and its rest energy by the sum line's photons; nature's proton mass is such a click mass, its quarks' rest no part of the comparison [1, the three masses of a body][3, only clicks against clicks] [1, the electron needs no band of its own; the rows against nature, (t)].

**The atom as a bound body.** An atom is a bound body of the holder of the sign: a nucleus, one quantum of its own family, writes the time angle  $\theta_0 = Z\alpha/r$  into the sign holder's row by its Wronskian, the Green's function 0.758 at the source Node; electrons, quanta of the matter family, read that angle into their rotation per proper interval; the standing records Rule3 makes in that angle are the atom's levels, Schrödinger's in the slow limit: with the nucleus's angle alone they have Bohr's form,  $E_n \propto 1/n^2$  with the radius  $a_0 = \sqrt{3}/(m^*Z\alpha)$ ,  $\bar{\lambda}_e/(Z\alpha)$  at nature's gap; the record also reads its own level, Hartree's self-term, repulsive and  $1/Z$  of the nuclear one, so the form is the law's for  $Z\alpha \gg \alpha$ , hydrogen-like ions converging to it as  $1/Z$ , and its consequence at  $Z = 1$  is a finding under a second derivation (S.46); its mass defect is the redshift of its electrons' rotation by the angle,  $(Z\alpha)^2/2$  of the electron's energy, the click layer's reading of a frequency; the orbital Zeeman splitting has  $g = 1$ . The atom is reflected on the board in Bohr's form, a supporting finding and not a claim of nature's hydrogen (DERIVED in form; CLICKS for the levels' ratios, the splitting and the redshift, GAMEBOARD for the records; S.46). No body of count below  $0.23 \Gamma$  localises at a Node, the law's threshold, so the compact spectrum begins at the Link's own energy,  $T$  per quantum, 465 TeV or more at the LHAASO bound, and no particle of nature is a compact bound body of the electron's quanta (COMPUTED; GAMEBOARD). The electron and the nucleons are quanta of their families: an electron is one quantum of count

$\sin \omega_e$ , ten orders below the compact threshold, and nothing smaller than a quantum clicks; mass is the family's gap,  $T \sin \omega_s$ , bound or free; what decays is a body, a quantum parted from it by a click at its front, and a quantum never breaks, having no part the click credits (DERIVED; CLICKS). The criterion, one line: what nature splits with energy is a body on the GameBoard, what no energy splits is a quantum of a family; the nucleus, the atom and the molecule are bodies by the one mechanism the law has for a body, a record standing in the well of a holder, the sign holder for the atom and the nuclear holder for the nucleus (ASSUMPTION) [1, the bound body is one Node; the nucleus][3, the electron and the nucleons are quanta, not pixels; no mass without a bound body; a bound body that breaks up is an atom that breaks up]. A nuclear holder the electron reads is excluded by the spectrum, hydrogen's 1S and muonic hydrogen (COMPUTED; CLICKS; S.46); so the nuclear holders of Section 10.3 are read by the nucleon families and not by the electron, a selective read declared. What the atom of the model does not give, each with its root: the nucleus from inside, its binding and its size 0.84 fm (the one denominator); spin one half only as a route (Section 10.3); the Lamb shift and the running of  $\alpha$  (no loops); Einstein's  $A$  coefficients and every lifetime (no spontaneous click); the families' masses, declared [1, the atom is a bound body of the holder of the sign].

**The lightest body, and the binding range.** Under the binding field a cloud stands only within the range  $R$ : the field holds no cloud whose Pekar radius  $5.25 \Gamma/C$  Links exceeds the range, so there is a lightest body,  $5.25 \Gamma/R$  quanta, 1,575 at  $R = 20$  with  $\Gamma = 6,000$ , the lightest baryon, and the electron is never a body. The cloud of 1,836 is not bound under  $R = 10$  or 15 and is bound under [2400, 2401],  $R = 20$ , its well 0.997 and its inertia 1.011 of its count and its fall 0.988 of a free packet's. So the file's pair is [2400, 2401], the range a world's choice that bound this cloud, no longer tied to the proton under the families' reading (Section 10.3) (COMPUTED, GAMEBOARD) [1, the rows against nature, (w)].

## 7.4 A body's clicks

What a detector reads of a body follows from the body's record at its Nodes, the readings of Section 4.3 and the paces, with no further line; every formula below is a derivation with its status and its fence, and none is a run's result [3, the mathematics of the bound bodies and how it bears on the clicks].

**A steady rotation writes no light.** A body at rest with the plane record  $\text{re} = A \cos \omega_b t$  and  $\text{im} = eA \sin \omega_b t$ ,  $e$  its ellipticity, has the Wronskian  $W = -eA^2 \sin \omega_b$  at every interval, so the holder of the sign's source is constant and its field stands at its rest: no travelling events, no click. A breathing body,  $A(t) = A_0(1 + \delta \cos \Omega t)$  over its Nodes, keeps each Node's Wronskian exactly, so a one-Node source cannot breathe; its light is written by the second moment of its charge, the retarded level  $\ell = (3/4\pi r)[S + \dot{Q}/2]$  with  $Q = \sum_i \sigma_i r_i^2$  and the monopole  $S$  constant by the continuity identity, so the light born carries the breathing's frequency  $\Omega$  and not the rotation  $\omega_b$ , and a far detector's click rate is that current's share through its front (S.42). So a symmetric body at rest clicks nowhere, and a body is seen by what it absorbs and by what it writes when it breathes or moves (DERIVED; the Wronskian and the source GAMEBOARD, the far detector's rate CLICKS) [1, light is born by the write].

**The invariant, and the count's drift.** For one Node of a body,  $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega(t) a_{\text{now}}$  with  $\omega$  changing slowly has the solution  $a_t = C \cos(\sum \omega + \phi)/\sqrt{\sin \omega(t)}$ , so  $A^2 \sin \omega_b$  is invariant, the discrete adiabatic invariant of Rule3's line, while the form  $D = A^2 \sin^2 \omega_b$  and the share are not; the recurrence stepped over 20,000 intervals keeps  $D/\sin \omega$  to  $4 \times 10^{-5}$ . The share in quanta at a Node,  $(\Gamma^2/p_i^2) A^2 \sin^2 \omega_b/T$  (Section 4.4), therefore moves under a moving well as

$$\frac{N(t_2)}{N(t_1)} = \frac{\sin \omega_b(t_2)}{\sin \omega_b(t_1)} \cdot \frac{p_i^2(t_1)}{p_i^2(t_2)}, \quad (13)$$

with no current through any Link: the first factor is the rotation's own change under the clock's pace, the second the Node's weight in the share (DERIVED for the invariant, to the first order in the slow variation, and for the drift; GAMEBOARD; S.14). The invariant is what nature calls the number of quanta,  $E/\omega$ , and the share is nature's energy; the same identity is Planck's relation on the lattice, share =  $I \sin \omega$  with  $I = A^2 \sin \omega/T$ , so that  $E = \hbar \omega$  holds with  $T$  for  $\hbar$  and  $\sin \omega$  in place of  $\omega$  (DERIVED; CLICKS) [1, the count is the record's share; the paces' own change moves the share with no current].

**The clock's click rate.** A bound body's rotation reads the content once: at rest  $\cos \omega_0(U) = \cos \omega_0 + 2gU$  to first order, with  $g = (\text{den} - \text{num})/\text{den}$ , so its rate falls by  $f(\omega_0)U$  with  $f(\omega_0) = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$ , 1.063 at [2, 3] and 1 as the gap closes,  $f = 1 + \omega_0^2/12 + \dots$ , the finite-rotation correction in  $\omega$  of a rate that is exactly  $N$  in  $2 \sin(\omega/2)$  (S.31). A light clock between two faces falls by  $2U$ , the Link's pace read twice (Section 10.2; S.15) (DERIVED, CLICKS).

**The resonance.** A bound record at the rotation  $\omega_b$  reads a light level  $b_L \cos \omega_L t$  into its content at the weight 1, the content  $x_0$  at its Node otherwise static. At  $k = 0$  the Node's line is  $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega(t) a_{\text{now}}$  with

$2 \cos \omega = 2 - 2g(1 - 2x + 2x^2)$ , so the light modulates the rotation by  $\epsilon = 4g(1 - 2x_0) b_L/\Gamma$ . The resonant part of the modulation at  $\omega_L = 2\omega_b$  grows the amplitude at  $\mu = \epsilon/(4 \sin \omega_b)$  per interval, the share at  $2\mu$ , inside the full width  $\epsilon/\sin \omega_b$  (S.16). The same modulation mixes the record's frequencies, so a body with two bound modes  $\omega_i < \omega_j$  is driven at the sum  $\omega_i + \omega_j$  and transfers between the two at the difference  $\omega_j - \omega_i$  at the rate  $\Omega_R = \epsilon_{ij}/(4\sqrt{\sin \omega_i \sin \omega_j})$ , Rabi's line (DERIVED, two derivations) [1, the two-mode line]. Under light's declared read, the holder's rotation of a plane record (Section 5.5), a constant angle is a gauge and the Link phases couple the modes through the inter-mode current, resonant at the difference line alone, Rabi's rate  $\Omega_R = \theta_L \omega_L |d|/2$  for a small line; the one-mode line at  $2\omega_b$  has no analogue there (DERIVED; S.43). The read's side is derived; the fraction absorbed, with the body's write back, is open (GAMEBOARD for the record, CLICKS for the body's share). [1, the rows against nature, (y)]

**A body's count in clicks.** A body's count is its share summed over its Nodes, what a full absorber credits, and under a moving well it drifts by Eq. (13) (DERIVED, CLICKS). The implementation's declared region credits the share, a bolometer's reading, Millikan's slope 0. Nature's detector is a body, and the model declares its click as one quantum's change in the absorbing body, the detector's fourth declaration beside its window, its region and its floor (ASSUMPTION) [3, the whole of a click is the detector's own transition]: the absorber answers at its own rotation by the two-mode line above (S.43), one transition per click, taking the share  $T(\sin \omega_j - \sin \omega_i)$ , one photon's share  $T \sin \omega_L$  to the order  $\omega_i \omega_j$ , below  $10^{-18}$  at nature's gap. So a body counts photons and reads  $\hbar \omega$  per click, Millikan's slope  $T$  (DERIVED; S.14). The share stays the conserved form and the draw's measure, so every ratio of the checks is unchanged (THEOREM); the implementation credits the share, the declared unit not yet implemented (CLICKS) [1, the count is the record's share].

## 7.5 Clusters: how they enter, and what each thing represents

**How a cluster enters.** The model writes no line for a cluster. One assumption names the universe it describes: every body is bound, no body is free, and a record that is in no body is a message between two clicks (ASSUMPTION). A cluster is bodies standing in one another's wells; between two bodies there is nothing but the click, the light of the holder of the sign and the two real fields [1, the bound body is one Node; the universe is bound; the algebra of clusters].

**The clicks between bodies are clicks in the one sense.** A body is a region of Nodes with a front, the Ports where its tail meets its neighbour's, so a body is an instrument in the meeting's form: the click between two bodies is the pair  $(R, t)$ , one whole quantum credited to the region of the taker at the interval  $t$ , drawn by the share  $Q_R/W_c$  of the inflow through its front on the forward orbit. A body's count is its share summed over its Nodes, and it moves only by the clicks at its front, up to the drift of Eq. (13) (DERIVED from the definitions; CLICKS) [3, the click's algebraic definition; the click is the meeting of the future with the past].

**What each thing represents in a cluster.** (i) *The three levels at a Node are the nesting.* The binding field's level at a Node is the matter's tail, the level of the holder of the sign is the charge's well and the massless field's level is gravity's well: the three kinds of binding by rank (DERIVED, GAMEBOARD). (ii) *Three kinds of binding, not three levels:* quanta into a body by the binding field's screened well, bodies into a charged cluster by the sign holder's field, charged clusters into a gravitational cluster by gravity's  $1/r$  well. (v) *The couplings are at the edge.* Every divisor is 1 but gravity's  $E_g$ , the universe of clusters carries  $\Gamma$ ,  $T$  and the two holders' weights and nothing else (ASSUMPTION for the divisors; GAMEBOARD) [1, the levels at a Node are the nesting; three kinds of binding, not three levels; the hierarchy is recursive; the click is the pointer; the observer is a cluster; the couplings are at the edge][3, what a body is; the rule's own universe is the universe].

**The clusters' cosmology.** The universe of the model is a finite ball of bodies flying through the static lattice, every body pulled by the others' inverse square at Kepler's  $G_K$  (Section 8, (j)). A shell of dust at the radius  $a$  feels the mass inside it, so  $\ddot{a} = -(4\pi G_K/3) \rho a$  and  $\dot{a}^2/a^2 = (8\pi G_K/3) \rho - \kappa/a^2$ , Friedmann's dust equations exactly for the ball's interior [48],  $\kappa > 0$  where the ball is bound; a receding emitter's redshift is  $1 + z = (1 + \beta)/\sqrt{1 - \beta^2}$  at  $c_m = c$ , nature's relativistic form to the second order (DERIVED, two derivations; CLICKS; S.29). Dust alone decelerates,  $q_0 = \Omega_m/2 > 0$ , where the supernovae read  $q_0 = -0.55 \pm 0.1$  [49][50][51], and no line of the rule as it stands turns the sign. One line writes a level without a body: a uniform write of the vacuum's row,  $-q_{\text{vac}}$  into the content at every Node each interval over a wall  $E_v$ , whose static rest is a bowl,  $U(r) = U_{\text{faces}} - |q_{\text{vac}}|(R^2 - r^2)/(2E_v \Gamma E_g)$ , every body falling outward: Newton's cosmological constant, under which the ball's equations are Friedmann's dust-plus- $\Lambda$  exactly,  $q_0 = \Omega_m/2 - \Omega_\Lambda = -0.55$  at  $\Omega_\Lambda = 0.7$ . Its one number is a declaration read from the supernovae's click. The finding is closed by an explicitly named hypothesis with one declared number, not derived (HYPOTHESIS; CLICKS; S.29) [1, the vacuum writes][1, the clusters' cosmology].

## 8 The forms of nature the rule arrives at

Each form below follows from the rule's coefficients, read by the geometric optics of a packet's band, before any run; the form of nature beside it is never an input. The coefficients contain no metric, no Lorentz transformation, no potential and no force, only the paces of Section 3.4, the clock composing by the rule and the Link the clock twice by the one assumption.  $U = c/\Gamma$  denotes the model's potential at the point named, the clock's share per level, and  $U_b$  its value at the impact parameter  $b$  [1, the rows against nature].

**Two potentials.** The forms are stated in two potentials. The model's own potential is the clock's share,  $U = c/\Gamma$ , the content  $x$  that the paces read; no instrument reads it. Every test of gravity takes  $GM$  from Kepler's law of the orbits of matter and  $c$  from light, so it reads  $U_K = GM_K/(c^2 r)$ . In the model a free packet of matter of the pair [num, den] falls in a content gradient at  $a = [(1 - \epsilon)/(3(1 - \epsilon/2))] |\nabla x|$  with  $\epsilon = (\text{den} - \text{num})/\text{den}$ , exactly from its band, Eq. (4), which with  $\cos \omega_0 = 1 - \epsilon$  and  $c^2 = 1/3$  is  $a = c^2 [2 \cos \omega_0 / (1 + \cos \omega_0)] |\nabla x|$ , so the orbits of matter measure

$$U_K = \frac{2 \cos \omega_0}{1 + \cos \omega_0} U, \quad 0.800 \text{ at } [2, 3], \quad 1 \text{ as the gap closes.} \quad (14)$$

Each form below, lettered as the law's rows are, is written first in  $U$ , where the composed paces give  $\gamma = \beta = 1$  exactly in the post-Newtonian words, the metric  $N^2 = e^{-2U}$  and  $h^2 = e^{2U}$ , and the redshift's factor  $f(\omega_0) = 1 + \alpha_U$ ,  $\alpha_U = 0.063$  at [2, 3], and then in  $U_K$ , where the same algebra gives  $\gamma_K = \text{den}/\text{num}$ ,  $\beta_K = (\text{den} + \text{num})/(2\text{num})$  and  $1 + \alpha_K = \tan \omega_0/\omega_0$ : 1.5, 1.25 and 1.329 at [2, 3], and 1 as the gap closes. The two columns are one algebra, and nature's tests read the second (DERIVED, CLICKS; S.18, S.19) [1, the rows against nature][3, general relativity in the law].

- (a) *The redshift* (DERIVED, CLICKS). Two clocks of one family at the levels  $c_1$  and  $c_2$  tick in the ratio  $1 - (U_1 - U_2)$  to first order; the rest rotation of a bound record, the clock this row names, shifts by  $f(\omega_0)U$  with  $f(\omega_0) = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$ , 1.063 at [2, 3] and 1 as the gap closes. Against the Newtonian potential the orbits measure it is  $z = (1 + \alpha_K)U_K$  with  $1 + \alpha_K = \tan \omega_0/\omega_0$ , 1.329 at [2, 3], where nature's clocks give  $|\alpha| < 10^{-4}$  [35]. Beside it: Pound and Rebka's line [52][35].
- (b) *The bending of light* (DERIVED, CLICKS). At a Node of content  $x$  light's index is  $n = e^{2x}$ ,  $1 + 2x$  to first order, because the Link's pace reads the content twice. Far from a body its field is the massless field's rest,  $L(r) = 3\sigma_{\text{tot}}/(4\pi r)$ , with  $\sigma_{\text{tot}}$  the body's wells per interval, and a beam passing at the closest distance  $b$  turns by the transverse gradient of the index integrated along the path:

$$\alpha = \int_{-\infty}^{\infty} \frac{\partial n}{\partial b} dl = \frac{3\sigma_{\text{tot}}}{2\pi\Gamma} \int_{-\infty}^{\infty} \frac{b dl}{(b^2 + l^2)^{3/2}} = \frac{3\sigma_{\text{tot}}}{\pi\Gamma b} = 4U_b, \quad U_b = \frac{3\sigma_{\text{tot}}}{4\pi\Gamma b}, \quad (15)$$

twice the angle  $2U_b$  that a Newtonian pull of the same potential gives. The factor two is the ruler going with the clock,  $hN = 1$ , the one assumption of Section 3.4. Against the Newtonian potential the orbits measure the bending is  $(1 + \gamma_K)2GM_K/(c^2 b)$  with  $\gamma_K = \text{den}/\text{num}$ , 1.5 at [2, 3], where Cassini's tracking gives  $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$  [35][53]. Beside it: Einstein's  $4GM/(c^2 b)$  [35].

- (c) *The delay of light* (DERIVED, CLICKS; S.20). Light's coordinate speed at the content  $U$  is  $e^{-2U}/\sqrt{3}$ , so its index is  $1 + 2U$  to first order, and a signal grazing a body at the impact parameter  $b$  from  $x_1$  to  $x_2$ , with  $U = \mu/r$ , is late by  $2\sqrt{3}\mu \ln(4x_1 x_2/b^2)$  intervals: Shapiro's delay with  $\gamma = 1$ . Beside it: Shapiro's delay [54].
- (d) *The perihelion* (DERIVED, CLICKS). By the same geometric optics the matter pair's periapsis advances by  $6\pi U_1/(a(1 - e^2))$  per orbit, Einstein's value, with the clock's second order; under the composed clock  $\beta = \gamma = 1$  exactly, Mercury's 43 arcseconds per century. Against the Newtonian potential the orbits measure the advance is  $(2 - \beta_K + 2\gamma_K)/3$  of Einstein's, 1.25 at [2, 3], where the planets' and the Moon's orbits give  $|\beta - 1| < 10^{-4}$  [35]. Beside it: Einstein's  $6\pi GM/(c^2 a(1 - e^2))$  [35].
- (e) *The moving clock: the band's kinetic scale and the fourth order* (DERIVED, CLICKS; S.21). The tick's factor of a packet moving at the velocity  $v$  is  $f = \Omega(k)/\omega_0$ , with  $\Omega(k) = \omega(k) - k\omega'(k)$  the rotation at the moving centre. From Rule3's exact band of a free record,  $\cos \omega = (\text{num}/(3\text{den}))(2 + \cos k)$ , the expansion in  $v$  gives  $f = 1 - v^2/(2c_m^2) - (2\omega_0/\sin 2\omega_0)v^4/(8c_m^4) + O(v^6)$  with  $c_m^2 = \omega_0/(3\tan \omega_0)$ , the band's kinetic scale, the second-order coefficient of the moving clock and no front. The fourth order is  $-(2\omega_0/\sin 2\omega_0)v^4/(8c_m^4)$ , its coefficient 1.6926 at [2, 3] and 1 as the gap closes, the one number of this row. Two speeds are to be told apart: the kinetic scale,  $c_m^2/c^2 = \omega_0/\tan \omega_0 = 1 - \omega_0^2/3$ , 13 percent below light's at the computed pair; the band's largest group speed,  $(\text{num}/\text{den})/\sqrt{3} = c \cos \omega_0$  at  $\mathbf{k} = (\pi/2, \pi/2, \pi/2)$ , 2/3 of light's at [2, 3] (S.21). At nature's matter pair the kinetic scale and the largest group speed are within  $10^{-19}$  of light's at the LHAASO bound,  $10^{-27}$  at the axis bound (Section 10.1), where nature bounds the difference of matter's and light's limiting speeds at  $10^{-15}$  to  $10^{-20}$  [55][56]. Beside it: Lorentz's factor, reached as  $c_m \rightarrow c$  [3, the click].

- (h) *De Broglie's relation* (DERIVED; GAMEBOARD for the band, CLICKS for the fringe spacing that reads it; S.22). A free record of a matter family has the band  $\cos \omega_b = (\text{num}/(3 \text{den}))(2 + \cos k)$ , the group velocity  $v = (\text{num}/(3 \text{den})) \sin k / \sin \omega_b$ , and to second order  $k = \omega_b v / c_m^2$  with  $c_m^2 = \omega_0 / (3 \tan \omega_0)$ , the band's kinetic scale. Beside it:  $\lambda = h/p$  with the rest rotation as the mass.
- (i) *The fall* (DERIVED, CLICKS). In a pace gradient a free packet or a cloud accelerates at  $a = \Delta |d\omega_0/dc| (c_+ - c_-)/D$  toward the well,  $\Delta$  its mode's top velocity and  $c_{\pm}$  the level at its two faces  $D$  Links apart. So bodies with the same  $\Delta d\omega_0/dc$  fall alike, and a cloud falls as a free quantum falls to 2 to 8 percent in the computed regime (Section 7.3, GAMEBOARD). Beside it: Galileo's fall and the equivalence principle [35]; and the binding by the holders of the content gravitates as in nature, its potential sitting in the record's count (S.38), two bodies of one count and different binding writing different wells and falling alike, while the seat of the electromagnetic binding energy is open (DERIVED for the holders of the content; CLICKS). Between families: a free packet of the pair [num, den] falls at  $c^2(1 - \tan^2(\omega_s/2)) \nabla U$  exactly, so two families of different gaps fall apart by  $(\omega_n^2 - \omega_p^2)/4$  at small gaps; with the proton and the neutron two families, beryllium against titanium gives  $\eta = 32.3 \omega_e^2$ , below  $3.9 \times 10^{-17}$  at the LHAASO bound, under nature's  $10^{-15}$  (DERIVED, CLICKS; S.47) [1, the rows against nature, (i)].
- (j) *Newton's pull and orbits* (DERIVED, CLICKS). The massless field rests at Poisson's solution with the sources and at Laplace's outside a body,  $c(r) = 3s/(4\pi r)$ , so the acceleration falls as  $1/r^2$  and a circular orbit's  $v^2$  as  $1/r$ . Beside it: Newton's pull and Kepler's fall-off.
- (m) *A free record keeps its wave number* (THEOREM, GAMEBOARD). Rule3 is invariant under translation, so a free record keeps its wave number and rotation exactly, and a radiating body's period is read unchanged at every distance. Beside it: Hubble's law as Doppler clicks of matter flying apart through the static lattice, with no stretching of space (Section 7.5).
- (p) *The lens: a content bends light and matter alike, by one formula* (DERIVED, two derivations; GAMEBOARD for the index, CLICKS for the focus; S.28). A stationary boundary conserves the rotation, so a record of the pair [num, den] entering a content  $U$  has its wave number inside fixed by  $1 - \cos k_{\text{in}} = [1 - \cos k + 3(\text{den} - \text{num})(1 - N^2)/\text{num}]/N^4$  with  $N = e^{-U}$  the clock, exact in  $k$  where the right side lies in  $[0, 2]$ . For light, num = den, the ratio is exact in the law's own variable,  $\sin(k_{\text{in}}/2) = \sin(k/2)/N^2$ , so the phase index  $k_{\text{in}}/k = (2/k) \arcsin[\sin(k/2)/N^2]$  is  $e^{2U}$ ,  $1/(1 - 2U)$  to first order, at long wavelength alone and carries the lattice's own chromatic term  $k^2(e^{6U} - e^{2U})/24$  at finite  $k$ ; the bending, the delay and the lens are read in the ray limit  $k \rightarrow 0$ . For slow matter,  $k, k_{\text{in}} \ll 1$ , the index is  $n_m^2 = k_{\text{in}}^2/k^2 = [1 + 6(\text{den} - \text{num})(1 - N^2)/(\text{num} k^2)]/N^4$ , growing as  $1/k^2$ , so one slab of content focuses slow matter far more than light: the fall of a body is the refraction of its own record, Newton's well the  $1/k^2$  term and the Link's double read on top, their ratio the Kepler column of Eq. (14). Beside it: nature's lens  $n = 1 + 2\Phi/c^2$  for light and  $n^2 = 1 + 2\Phi/v^2$  for a slow body, the law's two terms at a small gap [1, the content's index for every family; the rows against nature, (p)][3, one quantum through a lens, a beam splitter and an interferometer is the law's own].

**What goes in and what comes out.** The model puts in one rule, the paces' read under the one assumption of Section 3.4 and the run files' numbers. In the model's own potential the bending (b), the delay (c) and the perihelion (d) are Einstein's exactly,  $\beta = \gamma = 1$  in that column, light's ray metric; the post-Newtonian column at the computed pair departs with the gap. At the rule's own matter pair the model therefore does not describe the gravity of the solar system; the closing gap is nature's pair, where  $\gamma_K - 1$  is below  $10^{-18}$  to  $10^{-27}$  (Section 10.1), and whether the model's bodies stand at that pair is not shown. Read as a metric, the composed clock gives  $N^2 = e^{-2U}$  and the ruler  $h^2 = e^{2U}$ , against the isotropic Schwarzschild chart's  $1 - 2U + 2U^2$  and  $1 + 2U + 1.5U^2$ , the spatial second order 2 against 1.5. One click of the strong field reads the difference: light's index  $e^{2U}$  has its photon sphere where  $d(nr)/dr = 0$ , at  $r = 2m$  with  $U = m/r$ , so the capture radius of a black hole's shadow is  $b = 2em = 5.44m$  against Schwarzschild's  $3\sqrt{3}m = 5.20m$ , 4.6 percent larger, an explicit prediction (DERIVED, CLICKS; S.30). Beside it: the Event Horizon Telescope's shadows, Sgr A\*'s fractional deviation from Schwarzschild  $-0.08 \pm 0.09$  [57], 1.4 standard deviations from the prediction, and M87\*'s within 0.3 [58]. Multiplying Rule3's step by the record's conjugate gives an exact continuity identity, so the sign's density  $wW$  and its halved Link current  $-R_a J_a/2$  are one conserved four-current; and the time turn with the three Link phases,  $(\theta_0, \theta_a)$ , is one covector by minimal coupling, shifting  $(\omega, k)$  as the band's boost does. So the odd lines are sourced by the halved current over the same wall  $\text{den}T$  as the time level, no number matched, and a co-moving reader feels Coulomb's force over  $\gamma$ , 0.995 at  $0.1c$  and 0.866 at  $0.5c$  (DERIVED; CLICKS; S.33). A source and a reader of one family see one light speed, the band's own  $c_s^2 = \text{num}_s/(3 \text{den}_s)$ . At light's  $c$  the same to  $10^{-19}$  in nature at the LHAASO bound [1, the odd line's source is the sign's own current; the rows against nature, (b2)].

## 9 The two slits and Bell: the one experiment and the check, derived and read by the implementation's checks

The paper carries no experiment against nature: the results of the worlds below are derived from the rule and read by the implementation's checks (Section 3.7). The two slits is the one experiment, its fringe row fixed by the rule's line in real arithmetic and its arrival interval the lattice's own number; Bell's world is a check and not an experiment: its number is fixed by the declared credit and the parts' determinism. Nothing in a world is built to bring the right result [3, the paper; one experiment and one gate; nothing in an experiment brings the right result by construction].

### 9.1 The two slits: the world

The emitter is a declared packet of light. Its wavelength is 8 Links along  $+x$ ,  $k = \pi/4$ . Its levels are a raised-cosine packet of about ten quanta per Node,  $b = 1,328$ , over  $x = 4$  to 16, flat across  $y = 4$  to 44. The wall is the inner face at  $x = 20$  with two gaps of three Nodes,  $y = 17$  to 19 and 29 to 31,  $d = 12$  Links apart. The GameBoard is flat,  $45 \times 48 \times 1$  with the third axis folded, its  $y$  faces open and its  $x$  faces receding; the screen is the last column,  $x = 44$ ,  $L = 24$  Links from the wall. The screen is twelve declared detectors of four rows each; the run is 130 intervals, read over the intervals 45 to 130. A record uniform along the folded axis has the three-dimensional band at  $k_z = 0$ , so the flat GameBoard carries the world's band exactly (DERIVED, GAMEBOARD). On this world the law's line gives 278.0 quanta through the screen in the physical labels (277.8 in the implementation's labels), the peak of the screen's inflow at the interval 68.0 against 61.8 on the same world with the lattice refined fourfold (COMPUTED, CLICKS) [1, the rows against nature, (g)][3, the two slits; one experiment and one gate].

### 9.2 The two slits: what the model says it must give

Three readings of the one world, the Huygens prediction written before the run and the real line computed beside it. (a) *The fringe*. The far field's spacing is  $\lambda L/d = 16$  Links; at  $L = 2d$  the fringe is the near field's, with the minima at the rows 15 and 33 (DERIVED, CLICKS; S.23). Light's line at the vacuum's paces, stepped in real arithmetic on the declared world, gives 278.0 quanta through the screen (Section 9.1), per region as the script's table lists (COMPUTED, CLICKS; the script of Section 3.7); the Huygens sum of the two gaps gives 273 with the same central fringe. (b) *The clicks*. The screen's shares rounded to whole quanta are the expectation, and the row drawn by the shares at the declared seed is the clicks, with the draw's scatter  $\sqrt{Np(1-p)}$  per region; on the Huygens row at  $N = 273$  the central maximum stands above its two minima by 5.7 times its scatter. (c) *The lattice's own number*, the inflow peak's 68.0 against 61.8 (Section 9.1), which the click line resolves (COMPUTED, CLICKS). [1, the rows against nature, (g)][3, the click is the meeting of the future with the past].

### 9.3 Bell: the world

In the model's words, entanglement is one event with two trees. The pair is laid as the pair family, two real lines of light never summed at a Node, with equal parts. Each side's detector, a region at least half a wavelength across, reports the two parts' signed level sums at its ports and knows nothing of angles. The reader applies the declared setting, a pair of integers  $(p, q)$ , with the port coefficients  $e(+)= (p, q)$  and  $e(-)= (-q, p)$ , exactly orthogonal, so each side's marginal is one half for any setting. The settings of the Clauser-Horne-Shimony-Holt test are (1, 0), (1, 1), (12, 5) and (5, 12), at 0, 45, 22.62 and 67.38 degrees. The credit is the one form of the click: the joint share pairs the parts by their label across the two sides through the root,

$$J(p_A, p_B) = \left| \sum_k c_k e_k(p_A) \alpha_k(p_A) e_k(p_B) \beta_k(p_B) \right|^2, \quad c = (1, 1), \quad (16)$$

with  $\alpha_k$  and  $\beta_k$  the part  $k$ 's amplitude at the port on each side and  $e_k(p)$  the port's integer coefficient on the part  $k$ , accumulated over the window on both members of the level pair. The joint share is never the square of a local sum, which one detector could form alone and which gives  $S$  at most 2. One draw is made per pair: the first side from its marginal and the partner from the conditional (ASSUMPTION, CLICKS) [3, Bell in nature's own layout].

### 9.4 Bell: what the model says it must give

(1) Rule3 gives each part's amplitudes at the ports: the two parts of one beam, laid equal, are two records of one family stepped by the same line along the same path, identical bit for bit (THEOREM, GAMEBOARD). (2) The meeting with the linearity of Rule3 gives the joint amplitude: the realised click at A's port weights the parts by A's analyser, is read back to the root where the two parts were one event and forward to B, where B's analyser weights them, and the root sums the two branches,  $\sum_k c_k e_k(a) e_k(b) = \cos(a-b)$  for equal parts  $c = (1, 1)$  (DERIVED, CLICKS). (3) That the share is the density of the meeting, its square, is Born's rule for a pair [41]: nothing in Rule3 forces it; everything

after it, the fractions,  $S$ , GHZ's  $M$ , is derived in the algebra (ASSUMPTION, FITTED; CLICKS). From the three, with the pair's common phase  $\theta$  uniform over the pairs and A's + port at  $a$  and B's at  $b$ ,

$$\begin{aligned}
\text{the meeting:} & \quad P(++ ) = \frac{1}{2} \cos^2(a - b), \quad E(a, b) = \cos 2(a - b), \quad S = 2\sqrt{2}; \\
\text{a local credit by the shares:} & \quad P(++ ) = \frac{1}{4} + \frac{1}{8} \cos 2(a - b), \quad E = \frac{1}{2} \cos 2(a - b), \quad S = \sqrt{2}; \\
\text{a local credit by the sign:} & \quad E = 1 - 4|a - b|/\pi \quad (|a - b| \leq \pi/2), \quad S = 2; \\
\text{the credit by the contrast:} & \quad E = \cos 2(a - b) \text{ at the efficiency } 2/\pi, \quad S = 2\sqrt{2},
\end{aligned} \tag{17}$$

The second, third and fourth lines follow by averaging over the pair's phase  $\theta$  (DERIVED, CLICKS; S.10). So  $E(a, b) = \cos 2(a - b)$  under the meeting. For the declared pairs it is exact, Lagrange's identity of the reader on equal parts:  $E = ((p_A p_B + q_A q_B)^2 - (p_A q_B - q_A p_B)^2) / ((p_A^2 + q_A^2)(p_B^2 + q_B^2))$ , so at the four settings  $E = 119/169, -119/169, 120/169$  and  $120/169$  and  $S = 478/169 = 2.8284$  exactly, an efficiency of 1 by construction and the marginal one half on each side whatever the partner's setting (COMPUTED, CLICKS; S.10). For unequal parts,  $r$  their products' ratio at the ports, the same credit gives  $E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$  with  $\rho = 2r/(1 + r^2)$  and  $S = \sqrt{2}(1 + \rho)$  at the ideal angles (DERIVED; CLICKS). The maximum of  $S$  over the four settings for  $E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$  is  $2\sqrt{1 + \rho^2}$ , Tsirelson's bound [68]  $2\sqrt{2}$  at  $\rho = 1$  and never the algebraic 4, and the declared pairs'  $478/169 = 2.82840$  sits below  $2\sqrt{2} = 2.82843$  by the rational angles (DERIVED, CLICKS).

What the GameBoard gives without the declaration is the second and third lines of Eq. (17), two theorems about local credits on the same world (THEOREM, CLICKS): a credit of each side by the shares of its ports gives  $S = \sqrt{2}$ , and a credit by the sign of its larger port gives the sawtooth, whose sum is exactly 2, Bell's bound. Bell's value therefore sits in the credit rule, a declaration, and is no independent evidence that the rule accounts for entanglement. Nature beside the derivation:  $S = 2.42 \pm 0.20$  in the loophole-free test of 2015 that measured the sum [30], and the two photon tests of that year [31][32]. The same credit on four lines gives the three-quantum GHZ correlation [67]  $E_3 = \cos 2(a + b + c)$ , every pairwise  $E$  0 and Mermin's  $M = -4$  where every local credit gives at most 2 (DERIVED under the declared credit, CLICKS; S.40) [1, the GHZ gate]. No signal passes between the sides, each side's marginal being one half; no settings are chosen by the GameBoard's own past, so a superdeterministic world [59] is not this one [1, the rows against nature, (h)][3, the click is the meeting of the future with the past; Bell in nature's own layout].

## 10 Discussion: the constants, the calibration and the findings

### 10.1 The constants

The quantum of action is one click, whose energy is the quantum's norm  $T$ . Newton's constant is not declared: a body whose wells are  $s$  quanta per interval makes the massless field's rest  $3G(r)s$  at  $r$  Links over the field's divisor,  $3s/(4\pi r)$  far away, and the level slows the clock by the potential  $\Phi = -c/\Gamma = -U$ , Newton's sign, so (DERIVED, GAMEBOARD; S.25)

$$\frac{G_{\text{clock}}}{c^2} = \frac{3}{4\pi \Gamma E_g}, \text{ that is } G_{\text{clock}} = \frac{1}{4\pi \Gamma E_g} \text{ per quantum of well, in Links and intervals,} \tag{18}$$

a reading of  $\Gamma$ , the file's  $E_g$  and the lattice. Kepler's  $G$ , read from the orbits of matter as clicks, is  $[2 \cos \omega_0 / (1 + \cos \omega_0)] G_{\text{clock}}$  by Eq. (14), 0.8 of it at  $[2, 3]$  (DERIVED, CLICKS). The vacuum content is 0 in the file, a declaration no click bounds (Section 6.5), and the cosmological constant of Section 7.5 is the vacuum write's rate, not this rest [1, the constants].

**The fine-structure constant, a coefficient of the file; the units.** Under the holder's rotation (Section 5.5) the quantum of a family is one unit of the invariant,  $2A^2 \sin \omega_s = T$ , its energy  $T \sin \omega_s$  and its Wronskian  $A^2 \sin \omega_s = T/2$  for every family, the unit of charge, so charge universality is derived from the writes,  $|q_p| = |q_e|$  with nothing declared, nature's to  $10^{-21}$  (DERIVED; CLICKS; S.26). The holder of the sign carries one declared integer weight  $k$  in its write, the written level  $k$  times the plain one and the turn the level over  $\Gamma$ , as gravity's row carries  $E_g$ : one quantum writes  $1/(2E_s)$  per interval, the far angle is  $3k/(8\pi r \Gamma E_s)$ , and with de Broglie's relation and light's  $c$  the law's coupling between two quanta is  $\alpha_{\text{law}} = (3\sqrt{3}/8\pi) k/(\Gamma E_s)$ , independent of the family's gap (DERIVED under the rotation,  $k$  ASSUMPTION; CLICKS; S.26). The force on a body of  $n$  quanta and sense  $\pm$  is  $\pm n \nabla \theta$ , Coulomb's  $q_1 q_2$  with  $q = \pm ne$ ,  $e$  one quantum's Wronskian  $T/2$ ; the reader sources the holder it reads, so the pair is reciprocal (DERIVED; S.26, S.43). Nature's  $\alpha = 1/137.036$  fixes  $k/(\Gamma E_s) = 1/28.33$  and nothing else:  $\alpha$  is a coefficient of the file, read from one click as  $G$  is, neither derived nor a finding. The weight sits in the write,  $k = k_w k_r$  with the read's  $k_r = 1$ . A charged body and its light conserve by a line of the law: a plane record under the turn loses  $T \cos \omega_s \Delta \ell / \Gamma$  of form per quantum per step and the holder's own form gains  $(k_w/E_s) \Delta \ell$  under the write, an identity exact for every motion, so the two

agree if and only if  $E_s T \cos \omega_s = k_w \Gamma$ , the law's line (DERIVED for the identity, two derivations; the line ASSUMPTION; S.26); under it  $\alpha_{\text{law}} = (3\sqrt{3}/8\pi) T \cos \omega_s / \Gamma^2$ , at nature's gap  $T = \Gamma^2/28.33$ ,  $E_s$  a unit of the level and the file one integer fewer. The law has no theorem of energy conservation across families; this line is the first of the relations among the file's integers that is already a law. The two held rows are one kind of row, one kernel and the same acts of write and read; the sign holder reads a quantum's Wronskian  $T/2$ , its invariant, and the content holder its form,  $\sin \omega_s$ , its energy. So gravity's coupling between two quanta of the family  $s$  is  $\alpha_g(s) = (3\sqrt{3}/4\pi) \sin^2 \omega_s / (\Gamma E_g)$ , and the ratio of the two forces at that family is  $F_e/F_g = k E_g / (2E_s \sin^2 \omega_s)$ ; between two families the weights cancel,  $(F_e/F_g)_e / (F_e/F_g)_p = (m_p/m_e)^2$  with no number, nature's  $4.17 \times 10^{42} / 1.24 \times 10^{36} = 1836.15^2$  (DERIVED; CLICKS; S.44). On the board there is no  $10^{42}$ , only two rows, one kernel and two integers; the number appears when a detector compares the two forces on one quantum, and most of the electron's  $10^{42}$  is the smallness of its energy against its unit of charge,  $1/\sin^2 \omega_e > 8 \times 10^{17}$  at the LHAASO bound; the product of two declared weights explains nothing. The units are the electron's: one interval is  $\omega_e \hbar / (m_e c^2) = 1.288 \times 10^{-21} \text{ s} \times \omega_e$  and one Link  $\sqrt{3} \omega_e \bar{\lambda}_e = 6.69 \times 10^{-13} \text{ m} \times \omega_e$ . Light's band ends at  $\omega = \pi$  along the body diagonal, a highest photon energy  $1.605 \text{ MeV}/\omega_e$  (0.629 along an axis); LHAASO's photons of 1.4 PeV [60], the lattice's orientation unknown, give  $\omega_e < 1.1 \times 10^{-9}$ ,  $\text{den} > 1.5 \times 10^{18}$  and  $\gamma_K - 1 = \text{den}/\text{num} - 1 = \omega_e^2/2$  below  $6.6 \times 10^{-19}$ , and the Link's bound of Section 10.2 for a beam along an axis  $\omega_e < 2.1 \times 10^{-14}$  and  $\gamma_K - 1 < 10^{-27}$  (DERIVED, two derivations; CLICKS; S.27).  $\Gamma$  reads nothing of  $\omega_e$  or of  $\alpha$  and is bounded by nothing of the clocks: the content is an integer and the clock reads it as  $\Gamma(1 - 1/\Gamma)^c$ , so the finest level at one Node and one interval is  $1/\Gamma$ , but the write's carried remainder dithers a sub-unit level whose mean is the exact rest, which a clock averaging over its window reads, so the clocks bound the mean potential and not  $\Gamma$  (DERIVED, two derivations; S.26);  $\Gamma$  is free and large, bounded by nothing above and fixed by no line (CLICKS) [1, the law's alpha is a coefficient of the file; the units are the electron's and the band's top is the highest photon energy; what is open, item 19][3, the strong force's kernel derived: the colour coupling at the Link is the charge's alpha, no running, not the strong force].

## 10.2 Nature's numbers enter only as clicks

**What the lattice fixes with no number.** Before any file the lattice fixes: the geometry, a cube of Nodes with an integer state, six Ports and the 48 symmetries with the reflections, so parity is kept and there is one preferred frame, hence no  $g_{0a}$ , no frame dragging and  $\alpha_1 = -8$  (DERIVED; S.35, S.36); the dynamics, one rule per family, explicit, local and bijective with the remainder kept, so reversibility to the bit, no noise but the remainders' and no loops, hence no running of  $\alpha$  and no Lamb shift (THEOREM); the conservation, per family and no total across families: the form and so the energy per family, the Wronskian and so the charge under a static level, no spontaneous click and so no lifetime (DERIVED); the band, light at  $1/\sqrt{3}$  Link per interval under the causal bound 1, the anisotropy  $k^2/12$  at short wavelength, a gapped holder Yukawa's kernel with the massless constant (DERIVED); the paces, the exponential metric with  $\beta = \gamma = 1$ , the finest potential  $1/\Gamma$  and the frozen content  $\frac{1}{2} \ln(2\Gamma)$  (DERIVED under the one assumption); the holders, every held row Poisson's with the kernel  $3/(4\pi r)$ , its strength a declared weight and its range its gap (DERIVED); the click layer, the bilinear credit and so the angle it gives, one whole quantum per click and the draw by the shares (ASSUMPTION); and the scale, one scale  $\Gamma$  and pure numbers of order one, every large number a power of  $\Gamma$  or a declaration. In one sentence: the lattice is a machine of forms; it fixes every form and every ratio of order one, conserves, reverses and makes no noise; it fixes no strength, no scale and no large number; those are the file's [1, the properties of the lattice].

**The calibration series.** Every measured number of an experiment is first written as a click pattern on a declared detector against a declared clock. The families' pairs and the units  $\Gamma$  and  $T$  are fixed by that match and by nothing read off the GameBoard; the two ratios the model fixes without it are light's speed of  $1/\sqrt{3}$  Links per interval and the causal bound of 1 (ASSUMPTION, CLICKS) [1, the reading; nature's numbers enter the board only as clicks]. Every formula of Section 8 is derived in three steps: the law's line in the GameBoard's words; each word traded for a detector's report of it; and what remains, a relation among clicks alone, set beside nature's [3, only clicks against clicks]. No coefficient of the file is derived from the lattice; the lattice dictates the form in which each enters and reads none of them. Each is read from one click of nature, Table 5: every coefficient of the file is read by one click on a bound body; the lattice fixes the form of the reading; the file holds the integer. The table's last column places each in the three layers (Section 1):  $T$  and the detector's unit of credit are the click's, the gaps,  $k$ ,  $E_g$  and  $\Gamma$  the board's, read through bound bodies and given by no detector. A number from the lattice itself, a fixed point of the whole selecting  $\Gamma$  or a ratio of the weights, the law has none; if it exists it is in the cosmology, the vacuum's write, an explicitly named hypothesis for the next paper (HYPOTHESIS) [1, the calibration series]. One derived check beside the table, the two clocks: a cavity of bodies and an atom's beat shift alike by  $N$  at long wavelength, as nature's clocks do to  $10^{-5}$  [35], a cavity of declared faces at  $N^2$  (DERIVED; CLICKS; S.31).

**Table 5** The calibration series: each coefficient of the universe’s file, the one click of nature that reads it, and the layer it lives in (DERIVED for every form of reading, the integer the file’s). The lattice reads none of them.

| The coefficient                         | The one click that reads it                                                                                                          | The layer                                |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| $T$ , the unit of action                | Planck’s constant: the counter’s unit of credit and Millikan’s slope (Section 7.4)                                                   | the click                                |
| The Link and the interval               | $\omega_e$ ’s bounds from the highest photon, LHAASO’s 1.4 PeV, and light’s dispersion, MAGIC’s bound [61] (S.24, S.27)              | the board, through the electron’s family |
| $\Gamma$ , the board’s clock            | free and large, bounded by nothing of the clocks and fixed by no line (S.26)                                                         | the board                                |
| The gaps, one per family                | the masses: $\omega_e$ by the electron’s mass, $g_p/g_e = (m_p/m_e)^2$ under one denominator, read as click masses $T \sin \omega_s$ | the board, through bound bodies          |
| $E_g$ , gravity’s weight                | Newton’s $G$ , the fall and the orbits, Eq. (18)                                                                                     | the board, through the fall              |
| $k$ , the sign holder’s weight          | the fine-structure constant, the atom’s levels, $k/(\Gamma E_s) = 1/28.33$                                                           | the board, through the atom              |
| $c_{vac}$ , the vacuum content          | GW170817, one speed of light and gravity: 0                                                                                          | the board                                |
| The nuclear holders’ ranges and weights | the nuclear binding and the nuclei’s sizes, if the holders are admitted (Section 10.3; HYPOTHESIS)                                   | the board, through the nucleus           |

**Three predictions, untested.** Three numbers follow from the law’s lines where general relativity and quantum theory do not meet, each entered as an explicitly named hypothesis, read by no click yet (DERIVED, HYPOTHESIS; S.41). (1) *A clock in two places at once* (CLICKS). A bound body with a transition  $\Omega$ , laid as one record on two paths at two heights for a time  $T$ , has its beat scale as  $N = 1 - U$ ; so the two paths’ internal states differ at the recombination by the phase  $\Omega \Delta U T$  and the fringes’ visibility falls as  $|\cos(\Omega \Delta U T/2)|$ : 0.294 rad and 0.989 for a strontium clock at one metre for one second; the semiclassical expectation’s magnitude [62], unmeasured. (2) *The gravity of light* (GAMEBOARD). A light packet’s pull on a static body is its count’s,  $1 \times E/c^2$ , where general relativity’s pencil pulls at  $2 \times E/c^2$  [63]. (3) *The black hole’s quanta* (CLICKS). A collapse ends in a frozen clock from which no travelling event leaves (Section 7.2), so the law’s black hole radiates nothing,  $T_H = 0$  against Hawking’s  $6.2 \times 10^{-8}$  K for one solar mass [64]; its entropy its frozen Nodes’ levels, extensive [1, the rows against nature, (a2)].

### 10.3 Findings against nature and open questions

**GameBoard readings, not findings.** The rule’s own universe at [2, 3] is no finding against nature, which measures at its own pair (GAMEBOARD; Section 8, (e)). The masses are families. A cloud’s click mass is its count to  $(C/\Gamma)^2$  (DERIVED; GAMEBOARD), and nature’s mass ratios are not whole numbers,  $m_p/m_e = 1836.153$ ,  $m_\mu/m_e = 206.768$  and  $m_\pi/m_e = 273.13$ , so no particle of nature is a cluster of the electron’s quanta (DERIVED; CLICKS). So the masses are the families’ declared integers, and the proton is one quantum of its own family (ASSUMPTION); hydrogen’s clicks read it as one unit of  $W$  spread over 0.84 fm, four Compton lengths of its family (DERIVED, one derivation); Charge universality and the equivalence between families follow from the writes with no condition (DERIVED); if the families share one denominator, a declaration, the masses are the square roots of the gaps and the electron’s gap is a declared integer and not 1 (HYPOTHESIS). The model’s particle physics is families with declared masses and conversions by click, the conversion an explicitly named hypothesis [3, the proton’s two readings].

**Findings against nature.** The massless field’s travelling events carry the breathing polarisation and the lattice-aligned + and no  $\times$ , where nature’s gravitational waves carry + and  $\times$  at every orientation (CLICKS) [1, what is open, item 23]. Gravity’s readers are real records with no current’s read, so a gyroscope precesses by 0 from a rotating body’s motion, where Gravity Probe B measured  $37.2 \pm 7.2$  milliarcseconds per year against general relativity’s 39.2 [65] (DERIVED, CLICKS) [3, general relativity in the law; the current’s read]. A metric with no  $g_{0i}$  in its preferred frame has  $4\gamma + 4 + \alpha_1 = 0$ , so  $\alpha_1 = -8$  at  $\gamma = 1$  against nature’s  $|\alpha_1| < 4 \times 10^{-5}$  from the binary pulsar PSR J1738+0333 [66], a finding (DERIVED, CLICKS; S.35) [1, what is open, item 15]. The ball of dust alone decelerates against the supernovae’s  $q_0 = -0.55 \pm 0.1$ , an explicit finding closed by the vacuum’s write (Section 7.5; HYPOTHESIS).

**The lattice’s frame, and its mend as a hypothesis.** Why kinematic clicks hide the frame and gravitational clicks read it (DERIVED; CLICKS): at long wavelength every record’s band is a Lorentz form,  $\omega^2 = \omega_0^2 N^2 + c_m^2 k^2$ , so the moving clock’s  $\gamma$  follows and no Michelson-Morley shift appears. The content’s level is a Poisson rest in the lattice’s frame, so a moving mass writes the same bowl with no  $g_{0a}$  and no  $g_{ab}$  [1, the lattice’s frame is read by gravity alone, and its carrier is a pair of Links; what is open, item 15]. The mend, an explicitly named hypothesis with no line written (HYPOTHESIS): a carrier of a second kind, a level on a pair of Links, sourced by the mass current. Against it stands a theorem (S.36): no real, local, explicit, bijective lattice line on Rule3’s clock carries  $g_{0a}$ , the four ways tried closing as one. The carrier stays an explicitly named hypothesis, an object Rule3 would step and never a change of the line (Section 3.6) [3, nothing that is not Rule3 remains].

**The strong force, by derivation.** The strong force is two things: confinement, a theorem of the click with no coupling; and the nuclear binding, a declared pair of Yukawa holders read by the nucleon families alone, under which the nucleus's size, binding and its light are derived in form and its shells are not. Confinement is not a force: a click credits a whole record and never one line of it (Section 5.3), so nothing on the board is a quark to free (DERIVED; CLICKS; S.37). The nuclear binding is reached by no derived body of the law, the deuteron's 2.2 MeV against  $10^{-13}$  of it an explicit finding (CLICKS): the binding per quantum times a body's size in Links is bounded by  $0.24 CW/\Gamma$ , so a nucleon of  $10^{10}$  Links at the MAGIC Link ( $10^6$  at LHAASO's) bound at order one needs a weight  $W$  of  $10^{21}$  to  $10^{22}$  at  $\Gamma = 10^{14}$ , taken for illustration since no click fixes  $\Gamma$ , growing with  $\Gamma$  (THEOREM; GAMEBOARD; S.38). The nucleus is therefore a hypothesis under its own name: a bound body of nucleon quanta standing in the well of a nuclear holder, a gapped holder of the content, Yukawa's kernel with the massless constant (S.39), its range  $1/\kappa$  and its weight  $W_1$  declared, read by the nucleon families alone and by no other (the selective read of Section 7), and a second holder of shorter range and negative weight  $-W_2$  for the core, four declared integers per nucleus; the one-event record of  $A$  parts laid alike stays as the theorem of equal parts and not as the nucleus's reading (HYPOTHESIS). Derived under it, in form (DERIVED under the hypothesis, one derivation; S.45): the bound state's threshold, Yukawa's,  $W_1 \geq 1.173 \Gamma \kappa / \omega_p^3$ ; the binding at a body's size by the virial,  $b = (\lambda_s/R)^2/2$ , the deuteron's  $4.9 \times 10^{-3}$  and iron's  $1.6 \times 10^{-3}$  against nature's  $1.2 \times 10^{-3}$  and  $9.4 \times 10^{-3}$ , nature's order with no number; fusion as two nucleon quanta falling into the well, the pair breathing at the beat and writing its binding as light by the dipole (S.42); fission as the two-mode line under the rotation (S.43) from the bound mode to the continuum at  $\omega_L \geq B$ , the shape  $(\omega_L - B)^{3/2}/\omega_L^3$  of Bethe and Peierls [42] peaking at  $2B$ ; and the size law with the core. The findings: one well gives  $R \propto 1/A$  where nature's nuclei grow as  $A^{1/3}$  and saturate, the core holder the saturating declaration; the shells and the magic numbers (Pauli's exclusion) and the neutron's lifetime are explicit findings (CLICKS). [1, the strong force; the nucleus; what is open, item 1][3, is the strong force not events that do not split].

**The weak force.** The model has no electroweak theory; its weak interaction is the conversion, a click, an explicitly named hypothesis [1, the conversion]. Every rotation holder, massive or not, couples at  $\alpha$  at the Link, its gap its range and never its strength (DERIVED; GAMEBOARD; S.39); no  $G_F$  follows. A three-line record read as  $(\nu, e)$  under one  $O(3)$  has no fourth rotation, so no  $Z$ , where nature has the neutral current [69] and the  $Z$  [70]; parity is conserved, where nature's weak interaction violates it [71]; and the law has no spontaneous click, so the conversion has no rate and the neutron's 880 s no source: three explicit findings (CLICKS). What decays in the model is a body and never a quantum (Section 7), so the neutron's end is the conversion [3, the weak-force reading].

**The gaps in two columns.** Of the gaps, the first column is the world's, and the method invites another world; the second is the rule's, each by its theorem, what the rule says of nature. The world's:  $\alpha$ 's value through  $k$ ,  $G$  through  $E_g$ , the Link and the masses, the file's numbers read from nature; the hierarchy of gravity and electricity, the world's two weights, not the rule's; the spin holder and the click's unit, a world's declarations. The rule's, each with its root, none a number's fault [3, why the gaps]: the nuclear binding by any derived body (one denominator reads every level into the paces; THEOREM), closed only by the declared nuclear holders, whose size law is the finding; frame dragging,  $\alpha_1$  and the  $\times$  polarisation (no line carries  $g_{0a}$ ; THEOREM);  $\alpha$ 's running,  $1/137$  against  $1/128.9$  at the  $Z$ 's mass [70], the Lamb shift and the lifetimes (no loops, no spontaneous click); the anomaly  $g - 2$  and Pauli's statistics (no loops; no vector gives the exclusion); the  $Z$  and parity (the step commutes with the cube's reflections).

**Open.** Three routes stand to be checked in the implementation, explicitly named hypotheses (HYPOTHESIS; GAMEBOARD until run): the nucleus as a bound body in the declared nuclear holders, a light probe's scattering pattern its one click at a board's size, the hydrogen-like fall of the form factor against a one-event record's flat  $F = 1$ , and iron freezing its Node the size law's finding, nothing of it run; the weak interaction as the conversion at a click, the neutron's 880 s and  $G_F$  the click. An exactly conserved integer charge; gravity's divisor  $E_g$  (EXPERIMENT); the matter pair's gap; what fixes  $\Gamma$ ; a current's read for real records [1, what is open]. The relations among the file's integers are the next paper's program, each an explicitly named hypothesis tested by clicks: one denominator for all the families, the masses the square roots of the gaps; whether  $k$ ,  $E_g$ ,  $T$  and  $\Gamma$  are tied, the energy line  $E_s T \cos \omega_s = k_w \Gamma$  the first of them and already a law; the gaps' arithmetic; and the count of the law's free integers against nature's (HYPOTHESIS) [1, no free number but the mass].

## 11 Conclusion

One line of integer arithmetic at every Node of a cubic lattice is reversible to the bit and conserves a quadratic form with its current. A click is the meeting of the events run forward from an emitter, the future, with the same line read backward from the click, the past, at a declared detector, the choice among the meetings the one non-local element. Under the one assumption the static metric is the exponential metric, the weak-field forms Einstein's with  $\beta = \gamma = 1$

in the clock's potential and the shadow's capture radius  $2em$  a prediction; against Kepler's potential the matter band's factors reach 1 only as the family's gap closes. The paper's object is the method, on which the wave equation, the quantum of action and the metric are three readings of one line with nothing continuous beneath. Its formulas are examples, compared with nature at the stated order, Section 10.3 the list of what it does not yet give; the paper claims the method and not nature, every formula a supporting finding for it, and the method does not rest on any one of them. It is offered for the reader's judgement, test and further use. Between electricity and gravity the number is in the click, not on the board. Why nature is so, we do not claim to know.

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**Funding.** None. **Competing interests.** None. **Ethics approval and consent.** Not applicable. **Data and code availability.** The law's document, the implementation and the figures' scripts are developed in the public Git repository <https://github.com/Closer24/Universe24> and archived at Zenodo under the concept DOI 10.5281/zenodo.22738746 [40], MIT licence. **Author contributions.** The author laid down every postulate and decision of the model, directed the work and wrote the paper. **The use of a large language model.** The author used a large language model (Claude, Anthropic, through Claude Code, 2026) extensively as a tool, under the author's direction, for the software, the computations, the derivations and their checks, and the drafting of the text; the model is not an author, and the author reviewed every statement and takes full responsibility for the content.

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